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Busemann-Selberg Functions and Completeness for Dirichlet-Selberg domains in $\mathrm{SL}(n,\mathbb{R})/\mathrm{SO}(n)$

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Abstract We establish a general completeness criterion for Dirichlet-Selberg domains in the symmetric space $\mathrm{SL}(n,\mathbb{R})/\mathrm{SO}(n)$. By introducing **Busemann-Selberg functions** - which extend classical Busemann functions and capture asymptotic behavior toward the Satake boundary, we show that every gluing manifold or orbifold produced by Dirichlet-Selberg domain is complete. This result parallels the well-known hyperbolic case and ensures that the key completeness condition in Poincaré's Algorithm always holds in specific cases.

Keywords Geometric Group Theory · Symmetric Space · Satake Compactification · Poincaré's Fundamental Polyhedron Theorem · Dirichlet Domain

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1 Introduction

This paper is motivated by a semi-decidable algorithm based on Poincaré's Fundamental Polyhedron Theorem. The original version of Poincaré's Algorithm addresses the geometric finiteness of a given subgroup of $\mathrm{SO}^+(n,1)$. It was initially proposed by Riley [24] for the case $n = 3$ and was later generalized to higher dimensions by Epstein and Petronio [11]. Our research aims to generalize this algorithm to other Lie groups, particularly $\mathrm{SL}(n,\mathbb{R})$.

We begin by introducing the symmetric space corresponding to $\mathrm{SL}(n,\mathbb{R})$, namely $\mathrm{SL}(n,\mathbb{R})/\mathrm{SO}(n)$, and comparing it with hyperbolic space. Since the algorithm centers on the geometry of convex polytopes, we will also discuss the structure of convex polytopes in these settings.

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1.1 The Hyperbolic Spaces and the Symmetric Space $\mathrm{SL}(n,\mathbb{R})/\mathrm{SO}(n)$

Following the conventions in [17], we realize hyperbolic n-space via both the **hyperboloid model** in $\mathbb{R}^{n+1}$ and the **projective model** in $\mathbb{RP}^n$. In the hyperboloid model, a d-dimensional **plane** is determined by the intersection of $\mathbb{H}^n$ with a $(d+1)$-linear subspace; in the projective model, it is the intersection of $\mathbb{H}^n$ with a d-dimensional projective subspace.

Analogous to the hyperboloid and projective models of $\mathbb{H}^n$, we identify the space $\mathcal{X}_n = \mathrm{SL}(n,\mathbb{R})/\mathrm{SO}(n)$ through either a **hypersurface model**,

$$\mathcal{X}_{n,\mathrm{hyp}} = \{\mathbf{X} \in \mathrm{Sym}_n(\mathbb{R}) \mid \det(\mathbf{X}) = 1,\ \mathbf{X} > 0\},$$

or a **projective model**,

$$\mathcal{X}_{n,\mathrm{proj}} = \{[\mathbf{X}] \in \mathbb{P}(\mathrm{Sym}_n(\mathbb{R})) \mid \mathbf{X} > 0\}.$$

Here, $\mathrm{Sym}_n(\mathbb{R})$ denotes the vector space of $n \times n$ real symmetric matrices, and $\mathbf{X} > 0$ (or $\mathbf{X} \geq 0$) indicates that $\mathbf{X}$ is positive definite (or positive semi-definite). These models are standard in [10].

The hypersurface model of $\mathcal{X}_n$ is equipped with the metric tensor

$$\langle \mathbf{A}, \mathbf{B} \rangle_{\mathbf{X}} = 2n\mathrm{tr}(\mathbf{X}^{-1}\mathbf{A}\mathbf{X}^{-1}\mathbf{B}),\ \forall \mathbf{A}, \mathbf{B} \in T_{\mathbf{X}}\mathcal{X}_n.$$

The group $\mathrm{SL}(n,\mathbb{R})$ acts on $\mathcal{X}_n$ isometrically with respect to the metric tensor:

$$\mathrm{SL}(n,\mathbb{R}) \curvearrowright \mathcal{X}_n,\ g.\mathbf{X} = g^{\mathsf{T}}\mathbf{X}g.$$

The action is faithful when n is odd, and has a kernel of $\{\mathbf{I}_n\}$ when n is even, which makes $\mathrm{PSL}(n,\mathbb{R}) = \mathrm{Isom}_0(\mathcal{X}_n)$, the identity component of the isometry group, just as $\mathrm{SO}^+(n,1) = \mathrm{Isom}_0(\mathbb{H}^n)$.

The hypersurface and projective models allow us to define planes and half-spaces in $\mathcal{X}_n$ in a manner analogous to the hyperbolic case. We note, however, that such planes in $\mathcal{X}_n$ are not totally geodesic in general.

Following the conventions in [23], a **convex polytope** $P \subset \mathbb{H}^n$ is defined as the intersection of a locally finite collection of closed half-spaces. This notion adapts naturally to the definition of convex polytopes in $\mathcal{X}_n$. In both cases, we denote the set of facets (codimension-1 faces) by $\mathsf{S}(P)$ and the ridges (codimension-2 faces) by $\mathsf{R}(P)$. The set of all faces is denoted by $\mathsf{F}(P)$. We denote the plane of minimal dimension containing a face F by $\mathrm{span}(F)$.

While the hyperboloid model is suitable for describing hyperbolic convex polytopes, we transition to the projective model to define the **ideal boundary** $\partial_\infty\mathbb{H}^n$ and the **ideal closure** $\overline{\mathbb{H}^n}$ of hyperbolic n-space; these are, respectively, the boundary and closure of $\mathbb{H}^n$ in $\mathbb{RP}^n$. In the higher rank setting, the boundary and closure of $\mathcal{X}_n$ in $\mathbb{P}(\mathrm{Sym}_n(\mathbb{R}))$ are called its **standard Satake boundary**[3],

$$\partial_S\mathcal{X}_n = \{[\mathbf{X}] \in \mathbb{P}(\mathrm{Sym}_n(\mathbb{R})) \mid \mathbf{X} \geq 0,\ \mathbf{X} \not> 0\},$$

and **standard Satake compactification**,

$$\overline{\mathcal{X}_n}^S = \{[\mathbf{X}] \in \mathbb{P}(\mathrm{Sym}_n(\mathbb{R})) \mid \mathbf{X} \geq 0\}.$$

More generally, Satake compactifications [25] of Riemannian symmetric spaces correspond to irreducible projective representations with finite kernel of the associated Lie groups; see [15]. In our case, $\overline{\mathcal{X}_n}^S$ is associated with the standard representation of $\mathrm{SL}(n,\mathbb{R})$ on $\mathbb{R}^n$. When the context is clear, we omit the superscript S and simply write $\overline{\mathcal{X}}_n$, calling it the Satake compactification.

In the projective model of hyperbolic space, a convex polytope is the intersection of an $\mathbb{RP}^n$-polytope with the open region $\mathbb{H}^n$. This allows us to define the **ideal closure** of a convex polytope as its closure in $\mathbb{RP}^n$ and its **ideal boundary** as the intersection of its ideal closure with $\partial_\infty \mathbb{H}^n$. We say $a \in \partial_\infty \mathbb{H}^n$ is an **ideal vertex** of a convex polytope P, or that P is **incident with** a, if a is contained in the ideal closure $\overline{P}$.

By analogy, a convex polytope P in $\mathcal{X}_n$ can be viewed as the intersection of a $\mathbb{P}(\mathrm{Sym}_n(\mathbb{R}))$-polytope $\widehat{P}$ with the open region $\mathcal{X}_{n,\mathrm{proj}}$. Similar to the hyperbolic case, we define the **Satake compactification** $\overline{P}$ and the **Satake boundary** $\partial_S P$ for an $\mathcal{X}_n$-polytope P.

Unlike the hyperbolic scenario, the higher rank nature of the symmetric space $\mathcal{X}_n$ imparts a more intricate structure to the Satake boundary:

Proposition 11 ([3]) *The standard Satake compactification decomposes as*

$$\overline{\mathcal{X}_n} = \mathcal{X}_n \sqcup \bigsqcup_{k=1}^{n-1} \left(\mathrm{SL}(n,\mathbb{R})\mathcal{X}_k\right),$$

where each $\mathcal{X}_k = \mathrm{SL}(k,\mathbb{R})/\mathrm{SO}(k)$ *embeds into* $\partial_S \mathcal{X}_n$ *via*

$$\mathcal{X}_k \hookrightarrow \overline{\mathcal{X}_n},\ \mathbf{X} \mapsto \mathrm{diag}(\mathbf{X}, \mathbf{O}_{n-k}),$$

and $\mathrm{SL}(n,\mathbb{R})$ *acts by congruence on the set of positive semi-definite matrices.*

For any $g \in \mathrm{SL}(n,\mathbb{R})$ and $k = 1,\ldots,n-1$, we call the image $g \cdot \mathcal{X}_k \subset \partial_S \mathcal{X}_n$ in the aforementioned decomposition a **Satake boundary component** of $\mathcal{X}_n$. In the projective model, $g \cdot \mathcal{X}_k$ is identified with the set of positive semi-definite $n \times n$ matrices of rank k whose column vectors are contained in the k-subspace

$$\mathrm{span}(g \cdot \mathbf{e}_1, \ldots, g \cdot \mathbf{e}_k) \subset \mathbb{R}^n.$$

We denote by $\partial_S(V)$ the Satake boundary component corresponding to a linear subspace $V \subset \mathbb{R}^n$, and we say its **type** is $k = \dim V$. If a linear subspace is defined by a basis $(\mathbf{v}_1,\ldots,\mathbf{v}_k)$, we use the shorthand

$$\partial_S(\mathbf{v}_1,\ldots,\mathbf{v}_k) = \partial_S(\mathrm{span}(\mathbf{v}_1,\ldots,\mathbf{v}_k)).$$

Whenever $W \subset V$, the boundary component $\partial_S(W)$ lies in the boundary of $\partial_S(V)$; we denote this by $\partial_S(W) < \partial_S(V)$, and write $\partial_S(W) \leq \partial_S(V)$

if $W \subseteq V$. The **closure** of $\partial_S(V)$ is the disjoint union of all subordinate components:

$$\overline{\partial_S(V)} = \bigsqcup_{W \subseteq V} \partial_S(W).$$

Dually, the **star** of $\partial_S(V)$ consists of those components whose subspaces contain V:

$$\mathrm{st}(\partial_S(V)) = \bigsqcup_{U \supseteq V, U \subsetneq \mathbb{R}^n} \partial_S(U).$$

Both the Satake compactification and the polytope structure in $\mathcal{X}_n$ arise from the natural structure of the projective model. Consequently, the stratified structure of $\partial_S \mathcal{X}_n$ also applies to the Satake boundary of convex polytopes, particularly those of **finite volume**.

The finite-volume property for $\mathcal{X}_n$-polytopes with respect to the Riemannian volume form admits the following direct characterization:

Proposition 12 *Let $P \subset \mathcal{X}_{n,\mathrm{proj}}$ be a convex polytope, and let $\widehat{P} \subset \mathbb{P}(\mathrm{Sym}_n(\mathbb{R}))$ be the corresponding projective polytope. Then P has finite Riemannian $\mathcal{X}_n$-volume if and only if $\widehat{P} \subset \overline{\mathcal{X}_n}$. In other words, $\widehat{P}$ coincides with the Satake compactification $\overline{P}$.*

We adopt the characterization in Proposition 12 for the remainder of the paper. A proof of the proposition is given in Appendix A.

Since $\partial_S \mathcal{X}_n$ decomposes into boundary components indexed by subspaces of $\mathbb{R}^n$, a similar decomposition holds for $\partial_S P$:

Definition 11 *Let $P \subset \mathcal{X}_n$ be a finitely-sided convex polytope of finite volume. For each linear subspace $V \subset \mathbb{R}^n$, we define the **maximal Satake face***

$$\Phi_V = \partial_S P \cap \partial_S(V),$$

*provided the intersection is non-empty. The dimension of the subspace V is called the **type** of Φ_V.*

*We call any face of a maximal Satake face $\Phi \subset \partial_S P$ (including Φ itself) a **Satake face** of P, and denote the set of all Satake faces by $\mathsf{F}_S(P)$.*

*For any Satake face Φ, including the case when Φ is a Satake point, we denote the boundary component containing Φ by $\mathrm{supp}(\Phi)$. We call the **type** of Φ the same as the type of $\mathrm{supp}(\Phi)$.*

We denote the minimal-dimensional plane in $\mathrm{supp}(\Phi)$ containing Φ by $\mathrm{span}(\Phi)$.

Intuitively, Satake faces arise as ideal faces of $\mathcal{X}_n$-polytopes in the Satake compactification.

Generalizing the incidence between faces and ideal vertices in hyperbolic polytopes, we define the incidence between (ordinary) faces and Satake faces in $\mathcal{X}_n$-polytopes.

Definition 12 *In an $\mathcal{X}_n$ polytope P, we say that a face F is **incident with** a Satake face Φ if Φ is contained in the Satake compactification $\overline{F}$.*

*In particular, we say that F is incident with Φ **along** a boundary component Π if Φ is contained in $\overline{F} \cap \overline{\Pi}$. We say that F is incident with Φ **precisely along** Π if $\overline{\Phi} = \overline{F} \cap \overline{\Pi}$.*

It follows immediately that the closure of a maximal Satake face decomposes into smaller strata:

$$\overline{\Phi_V} = \bigsqcup_{W \subset V} \Phi_W.$$

Furthermore, the Satake boundary of any finite-volume, finitely-sided polytope admits a natural combinatorial description:

Proposition 13 *Let $P \subset \mathcal{X}_n$ be a finitely-sided convex polytope of finite volume, with Satake compactification $\overline{P}$. Then for each maximal Satake face $\Phi_V \subset \partial_S P$, its closure $\overline{\Phi_V}$ is a face of the projective polytope $\overline{P}$.*

The proof is included in Appendix B.

This leads to the following:

Corollary 11 *Let $P \subset \mathcal{X}_n$ be a finitely-sided, finite-volume convex polytope. Then:*

- *There are only finitely many subspaces $V \subset \mathbb{R}^n$ for which $\Phi_V \neq \varnothing$. Equivalently,*
$$\partial_S P = \bigsqcup_{V \in \mathsf{V}} \Phi_V,$$
is a disjoint union over a finite collection V of $\mathbb{R}^n$-subspaces.
- *Upon identifying a Satake boundary component $\partial_S(V)$ of type k with $\mathcal{X}_k$, the maximal Satake face Φ_V (if non-empty) is identified with a finitely-sided $\mathcal{X}_k$-convex polytope of finite volume.*
- *Under this same identification, the maximal Satake faces Φ_W with $W \subsetneq V$ correspond to the maximal Satake faces of Φ_V (viewed as an $\mathcal{X}_k$-polytope).*

1.2 Poincaré's Algorithm: Exactness, Facet Pairing, and Ridge Cycles

Before discussing the Poincaré Fundamental Polyhedron Theorem and Poincaré's Algorithm, we must introduce several preliminary concepts following the treatment in [23].

A convex polytope P is **exact** if every facet $F \in \mathsf{S}(P)$ is associated with an isometry $g_F \in \mathrm{SO}^+(n,1)$, called a **facet pairing transformation**, that identifies F with a facet $F' = g_F^{-1} F$ (where F and F' are not necessarily distinct). A standard example of this construction is the Dirichlet domain $\mathrm{D}(x, \Gamma)$ for a discrete subgroup Γ, where the isometries $g \in \Gamma$ serve as the canonical pairing transformations for the facets contained in the bisectors $\mathrm{Bis}(x, g.x)$.

The set of transformations $\Phi = \{g_F\}$ defines a **facet pairing** on P, inducing an equivalence relation $\sim$ on its points. This identifies paired points on the facets to form the quotient space $M = P/\sim$.

The Poincaré Fundamental Polyhedron Theorem provides conditions under which an exact convex polytope P serves as a fundamental domain for a discrete subgroup Γ. This requires tracking the behavior of P around its codimension-2 faces, or **ridges**.

For any ridge $r \in \mathsf{R}(P)$ and an interior point $x \in r$, we consider its **cycle**: a sequence $\{x_i\}_{i=1}^m$ of points in P such that $x_1 = x$, and each x_i is paired with x_{i+1} by a facet pairing transformation $g_i = g_{F_i}^{-1}$ mapping the ridge $r_i \ni x_i$ to $r_{i+1} \ni x_{i+1}$, for $i = 1, \ldots, m$ (with indices taken modulo m). Each cycle is associated with a **dihedral angle sum**:

$$\theta[x] = \sum_{i=1}^{m} \theta(x_i),$$

representing the total angle accumulated by the facets meeting at that ridge. A facet pairing Φ is called **proper** if each equivalence class of points related by Φ is finite, and the dihedral angle sum of each ridge-cycle is a submultiple of 2π.

If P is an exact convex polytope and Φ is a proper facet pairing, then the quotient space $M = P/\sim$, formed by **gluing** the facets via Φ, is a hyperbolic orbifold ([23], Theorem 13.4.2). Furthermore, if M is a complete space, the Poincaré Fundamental Polyhedron Theorem asserts that the group $\Gamma = \langle \Phi \rangle$ generated by the facet pairing transformations is a discrete subgroup of $\mathrm{SO}^+(n,1)$, with P as its fundamental domain.

The concepts and assertions presented above for $\mathbb{H}^n$-polytopes under the $\mathrm{SO}^+(n,1)$-action generalize to $\mathcal{X}_n$-polytopes under the $\mathrm{SL}(n,\mathbb{R})$-action as well. A primary concern involves the dihedral angle sum, since the dihedral angles between two facets of an $\mathcal{X}_n$-polytope depend on the choice of the base point. This dependency is explored further in Subsection 3.3. Nevertheless, given a cycle $\{\mathbf{X}_i\}_{i=1}^m$ where each $\mathbf{X}_i$ is a point in $\mathcal{X}_n$, the Riemannian angle $\theta(\mathbf{X}_i)$ is unambiguous; consequently, the dihedral angle sums

$$\theta[\mathbf{X}] = \sum_{i=1}^{m} \theta(\mathbf{X}_i),$$

as well as the properness of the facet pairings, remain valid. In particular, an exact $\mathcal{X}_n$-polytope P with a proper facet pairing Φ glues into a quotient space $M = P/\sim$, with an $(\mathcal{X}_n, \mathrm{PSL}(n,\mathbb{R}))$-atlas coming from glued copies of neighborhoods. This structure makes M an $(\mathcal{X}_n, \mathrm{PSL}(n,\mathbb{R}))$-orbifold. The general concepts of (X,G)-atlas and (X,G)-manifolds and orbifolds refer to [23], Sections 8.3 and 13.2.

In [8], we introduced an alternative to Riemannian dihedral angles—namely, an angle-like function that does not depend on the choice of the base point. This approach applies to generic pairs of hyperplanes, simplifying the implementation of Poincaré's Algorithm.

There are various choices for candidate polytopes used in Poincaré's algorithm. In [24], Riley employed "trial" Ford domains for subgroups of $\mathrm{PSL}(2,\mathbb{C})$.

As suggested in [18], another option is a generalization of **Dirichlet domains**, where subsets are considered in place of subgroups of $\mathrm{SO}^+(n,1)$.

Definition 13 *For a point x in hyperbolic n-space $\mathbb{H}^n$ and a discrete subset Γ_0 of the Lie group $\mathrm{SO}^+(n,1)$, the **(pre-) Dirichlet domain** for Γ_0 centered at x is defined as*

$$\mathrm{D}(x,\Gamma_0) = \{y \in \mathbb{H}^n \mid d(g.x,y) \geq d(x,y),\ \forall g \in \Gamma_0\}.$$

By contrast, classical Dirichlet domains in $\mathcal{X}_n$ are non-convex and thus unsuitable for Poincaré's algorithm. Kapovich [18] considered an $\mathrm{SL}(n,\mathbb{R})$-invariant proposed by Selberg [26] as a substitute for the Riemannian distance and defined Dirichlet–Selberg domains:

Definition 14 *For $\mathbf{X},\mathbf{Y} \in \mathcal{X}_n$, the **Selberg invariant** from $\mathbf{X}$ to $\mathbf{Y}$ is defined as*

$$\mathfrak{s}(\mathbf{X},\mathbf{Y}) = \mathrm{tr}(\mathbf{X}^{-1}\mathbf{Y}).$$

*For a point $\mathbf{X} \in \mathcal{X}_n$ and a discrete subset $\Gamma_0 \subset \mathrm{SL}(n,\mathbb{R})$, the **(pre-) Dirichlet–Selberg Domain** for Γ_0 centered at $\mathbf{X}$ is defined as*

$$\mathrm{DS}(\mathbf{X},\Gamma_0) = \{\mathbf{Y} \in \mathcal{X}_n \,|\, \mathfrak{s}(g.\mathbf{X},\mathbf{Y}) \geq \mathfrak{s}(\mathbf{X},\mathbf{Y}),\ \forall g \in \Gamma_0\}.$$

Without causing confusion, we may call pre-Dirichlet domains and pre-Dirichlet-Selberg domains just as Dirichlet domains and Dirichlet-Selberg domains for convenience.

For any discrete subgroup $\Gamma < \mathrm{SL}(n,\mathbb{R})$ and $\mathbf{X} \in \mathcal{X}_n$ with $\mathrm{Stab}_\Gamma(\mathbf{X}) = \mathbf{1}$, the Dirichlet–Selberg domain $\mathrm{DS}(\mathbf{X},\Gamma)$ is a fundamental domain of Γ [18]. Furthermore, pre-Dirichlet–Selberg domains are $\mathcal{X}_n$-polytopes: they are bounded by **(Selberg) bisectors**,

$$\mathrm{Bis}(\mathbf{X},g.\mathbf{X}) = \{\mathbf{Y} \in \mathcal{X}_n \mid \mathfrak{s}(\mathbf{X},\mathbf{Y}) = \mathfrak{s}(g.\mathbf{X},\mathbf{Y})\},$$

which are hyperplanes in $\mathcal{X}_n$ consisting of points $\mathbf{Y} \in \mathcal{X}_n$ satisfying the linear equation

$$\mathrm{tr}\left(Y \cdot \left(\mathbf{X}^{-1} - (g.\mathbf{X})^{-1}\right)\right) = 0.$$

Adopting the generalized Dirichlet and Dirichlet–Selberg domain constructions, we outline Poincaré's algorithm for $\mathrm{SO}^+(n,1)$ and $\mathrm{SL}(n,\mathbb{R})$. The algorithm for $\mathrm{SL}(n,\mathbb{R})$ was first discussed by Kapovich ([18], following Theorem 11.4) and explicitly stated in [8]. In the following, let X denote the symmetric space ($\mathbb{H}^n$ or $\mathcal{X}_n$) and G the corresponding isometry group ($\mathrm{SO}^+(n,1)$ or $\mathrm{SL}(n,\mathbb{R})$).

Algorithm 11 (The Poincaré's Algorithm) *Input: Generators $g_1,\ldots,g_m \in G$ and a base point $x \in X$.*

Output: `true` and a finite presentation of $\Gamma = \langle g_1,\ldots,g_m\rangle$ if the Dirichlet domain $\mathrm{D}(x,\Gamma)$ is finitely sided. Otherwise, the algorithm may not terminate.

1. *Set $l = 1$ and compute the finite subset $\Gamma_l \subset \Gamma$ consisting of elements represented by words of length $\leq l$ in the generators g_i and g_i^{-1}.*

2. *Compute the face poset of the Dirichlet domain* $\mathrm{D}(x, \Gamma_l)$ *(or* $\mathrm{DS}(x, \Gamma_l)$ *in the higher rank setting), which defines a finitely sided polytope in* X*.*
3. *Utilizing this face poset data, check if* $\mathrm{D}(x, \Gamma_l)$ *satisfies the following conditions:*
   (a) *Verify that* $\mathrm{D}(x, \Gamma_l)$ *is an* ***exact convex polytope****. Specifically, for each* $w \in \Gamma_l$*, confirm that the isometry* w *pairs the two facets contained in* $\mathrm{Bis}(x, w.x)$ *and* $\mathrm{Bis}(x, w^{-1}.x)$*, provided these facets exist.*
   (b) *Verify that the canonical facet pairing for* $\mathrm{D}(x, \Gamma_l)$ *is* ***proper*** *by computing the dihedral angle sums of the ridge cycles, as previously described.*
   (c) *Verify that each generator* g_i *can be expressed as a product of the facet-pairing transformations of* $\mathrm{D}(x, \Gamma_l)$*. This is done by determining the word in the facet-pairing transformations that maps* $g_i.x$ *into the interior of* $\mathrm{D}(x, \Gamma_l)$ *and comparing that word to* g_i *as matrices in* G*. (See [24], the last part of Section 1; Riley's method extends to* $\mathrm{SL}(n, \mathbb{R})$ *as a matrix group).*
4. *If any of these conditions are not met, increment* l *by* 1 *and repeat the process.*
5. *If all conditions are satisfied, verify if the quotient space of* $\mathrm{D}(x, \Gamma_l)$ *is complete (discussed in the following subsection). If so, by the Poincaré Fundamental Polyhedron Theorem,* $\mathrm{D}(x, \Gamma_l)$ *is a fundamental domain for* Γ*. Return* `true` *to indicate* Γ *is discrete, and output the finite presentation derived from the ridge cycles of* $\mathrm{D}(x, \Gamma_l)$ *[23].*

We have implemented this algorithm in *Python* [9]. The program includes several key functions: `compute_selberg_domain` constructs the Dirichlet–Selberg domain $\mathrm{DS}(\mathbf{X}, \Gamma_l)$ for a given center $\mathbf{X}$, generators g_i, and maximal length l; `polytope_exactness` checks if the resulting domain is exact under the canonical facet pairings. For exact domains, `compute_ridge_cycles` identifies the ridge cycles, while `compute_angle_sum` calculates the angle sum for each cycle to determine if the pairing is proper. Finally, `is_word_recovered` verifies whether the original generators are recovered by the facet-pairing transformations.

While Poincaré's Algorithm is effective for discrete subgroups of $\mathrm{PSL}(2, \mathbb{R})$ and $\mathrm{PSL}(2, \mathbb{C})$, it may not terminate in dimensions 4 or higher. For instance, Bowditch [4] constructed a geometrically finite subgroup of $\mathrm{SO}^+(4, 1)$ that lacks a finitely sided Dirichlet domain. In such cases, the algorithm fails to construct a finitely sided fundamental polytope and cannot determine the group's discreteness or presentation.

This phenomenon persists in the higher rank scenario. In [8], we identified multiple instances of elementary subgroups of $\mathrm{SL}(3, \mathbb{R})$ that do not admit finitely sided Dirichlet–Selberg domains. Additionally, Davalo and Riestenberg [7] demonstrated that uniform lattices in $\mathrm{SO}(n-1, 1)$, when viewed as subgroups of $\mathrm{SL}(n, \mathbb{R})$ via the canonical inclusion

$$\mathrm{SO}(n-1, 1) \hookrightarrow \mathrm{SL}(n, \mathbb{R}),$$

do not admit finitely sided Dirichlet–Selberg domains for any choice of center. This provides a negative answer to Kapovich's question regarding whether Anosov subgroups always admit finitely sided Dirichlet–Selberg domains [18].

A sufficient condition for the effectiveness of Poincaré's Algorithm is the uniform lattice property, as Dirichlet-Selberg domains of these subgroups are compact thus finitely-sided. By contrast, it is unclear whether every nonuniform lattice have finitely-sided Dirichlet-Selberg domains [18].

As a more sophisticated sufficient condition, Davalo and Riestenberg showed that $|\log \omega_i|$-undistortedness for $\mathrm{SL}(2n, \mathbb{R})$-subgroups admit finitely sided Dirichlet–Selberg domains for every choice of center, ensuring the algorithm terminates in finitely many steps[7]. Examples include the Schottky groups in $\mathrm{SL}(2n, \mathbb{R})$ we constructed in [8]. However, this property does not extend to non-virtually cyclic subgroups of $\mathrm{SL}(2n-1, \mathbb{R})$.

1.3 The Completeness Condition

Another significant aspect of Poincaré's algorithm is the completeness condition in Step 5, which is necessary for applying the Poincaré Fundamental Polyhedron Theorem.

In the context of hyperbolic 3-space, this condition is closely related to **hyperbolic Dehn filling** [27]. Consider, for instance, the **Meyerhoff manifold** [21], which arises from completing an incomplete gluing of a certain ideal triangular bipyramid. This manifold corresponds to the $(5, 1)$-Dehn filling of the **figure-eight knot complement**. While the ridge cycles of the Meyerhoff manifold provide relators for the facet pairings that combinatorially match those of the figure-eight knot group, additional relators arise due to the Dehn filling condition [22]. Consequently, the Poincaré Fundamental Polyhedron Theorem cannot fully recover the group presentation generated by the aforementioned Meyerhoff facet-pairing transformations.

Fortunately, for hyperbolic Dirichlet domains, the completeness condition is not a concern, as it is always satisfied for exact convex polytopes with a proper canonical facet pairing. The guaranteed satisfaction of the completeness condition can be explained through **Busemann functions** and **ideal vertex cycles**.

Following the convention in [5], for any $a \in \partial_\infty \mathbb{H}^n$ and $x \in \mathbb{H}^n$, the **Busemann function** $\mathrm{b}_{a,x} : \mathbb{H}^n \to \mathbb{R}$ is defined as the limit

$$\mathrm{b}_{a,x}(y) = \lim_{t \to \infty} \left(d(c(t), y) - t \right),$$

where $c : \mathbb{R}_{\geq 0} \to \mathbb{H}^n$ is the geodesic ray from $x = c(0)$ to $a = c(\infty)$. Note that changing the reference point x only shifts the Busemann function by a constant. Geometrically, the level sets of $\mathrm{b}_{a,x}$ are **horospheres** based at a.

For our purposes, the Busemann function $\mathrm{b}_{a,x}$ is significant due to its asymptotic behavior: it decreases to $-\infty$ only along geodesic rays asymptotic to a, and increases to $+\infty$ along geodesic rays asymptotic to any $b \neq a$ in $\partial_\infty \mathbb{H}^n$.

Ideal vertex cycles generalize the notion of ridge cycles and are closely related to the completeness condition. This concept was implied in [11] (Theorem 6.3) and explicitly addressed in [18] (discussion preceding Theorem 11.4):

Definition 15 *Let a be an ideal vertex of an exact convex polytope $P \subset \mathbb{H}^n$. A **cycle** of a is a sequence $\{a_i\}_{i=1}^m$ of ideal vertices of P such that $a_1 = a$, and each a_i is paired with a_{i+1} (with indices taken modulo m). That is, there exists a facet $F_i \in \mathsf{S}(P)$ incident with a_i such that the paired face F_i' is incident with a_{i+1}, with $g_{F_i'} \cdot a_i = a_{i+1}$.*

Define the product

$$w = g_{F_m'} g_{F_{m-1}'} \cdots g_{F_1'},$$

*as the **holonomy** for the ideal vertex cycle.*

Regarding the ideal vertex cycles of exact Dirichlet domains (defined with respect to subsets of $\mathrm{SO}^+(n,1)$), the Busemann function exhibits the following invariance property:

Lemma 11 ([18], discussion following Conjecture 11.5) *Suppose that $P = \mathrm{D}(x, \Gamma_0)$ is an exact Dirichlet domain centered at $x \in \mathbb{H}^n$ for a finite subset $\Gamma_0 \subset \mathrm{SO}^+(n,1)$. Let $a \in \partial\mathbb{H}^n$ be an ideal vertex of P with an ideal vertex cycle $\{a_i\}_{i=1}^m$, corresponding to the facet-pairing transformations $g_{F_i'}$, where $g_{F_i'} \cdot a_i = a_{i+1}$ for $i = 1, \ldots, m$ (taken modulo m). Then the holonomy w for the cycle is either elliptic or parabolic, fixes the ideal vertex a, and preserves the Busemann function based at a.*

For each ideal vertex a_i, a corresponding horosphere that intersects only the faces incident with a_i bounds a **cusp region** in P [23]. The aforementioned preservation of the Busemann function ensures that Cauchy sequences in the cusp region of the quotient $P/\!\sim$ remain bounded away from the visual boundary, thereby guaranteeing the completeness condition in Step (5) of Poincaré's algorithm:

Proposition 14 ([18], the discussion after Conjecture 11.5) *Let $P = \mathrm{D}(x, \Gamma_l)$ be an exact, finitely sided Dirichlet domain in $\mathbb{H}^n$ with a proper canonical facet pairing. Then the quotient space $M = P/\!\sim$ is complete. In particular, P is a fundamental domain for the subgroup generated by its facet pairings.*

Kapovich conjectured that an analogous property holds for Dirichlet–Selberg domains:

Conjecture 11 ([18], Conjecture 11.5) *Let $P = \mathrm{DS}(\mathbf{X}, \Gamma_l)$ be a finitely sided Dirichlet–Selberg domain in $\mathcal{X}_n$ with a proper canonical facet pairing. Then the quotient space $M = P/\!\sim$ is complete. In particular, P is a fundamental domain for the subgroup generated by its facet pairings.*

We expect to rely on similar, but more elaborated strategy for proving this conjecture. As one of the key ingredient, we can generalize ideal vertex cycles to **Satake cycles** in the higher rank settings.

Definition 16 *Let Φ be a Satake face of an exact convex polytope $P \subset \mathcal{X}_n$. A **cycle** of Φ is a sequence $\{\Phi_i\}_{i=1}^m$ of Satake faces of P such that $\Phi_1 = \Phi$, and each Φ_i is paired with Φ_{i+1} (indices taken modulo m): there exists facets $F_i \in \mathsf{S}(P)$ incident with Φ_i, such that F_i' is incident with Φ_{i+1}, with $g_{F_i'}.\Phi_i = \Phi_{i+1}$.*

Define the product

$$w = g_{F_1'} g_{F_2'} \cdots g_{F_m'},$$

*as the **holonomy** for the Satake cycle.*

A generalization of Lemma 11 will be discussed in Subsection 3.2.

For the other ingredient, one can construct horospheres on $\mathcal{X}_n$ that are preserved by a certain word $w \in \mathrm{SL}(n,\mathbb{R})$ in the facet-pairing transformations, similar to the conclusion of Lemma 11 for the hyperbolic case (see Subsection 2.1). However, a key difference in the higher rank setting is that this preservation does not guarantee the strict parabolicity of w. Consequently, we cannot deduce completeness through this property alone.

To explore further properties of Dirichlet–Selberg domains beyond this straightforward preservation, we introduce **Busemann functions** of different types, which appear to be effective - at least for Dirichlet–Selberg domains of finite volume. These domains correspond to **lattices** in $\mathrm{SL}(n,\mathbb{R})$, which play a central role among the discrete subgroups of $\mathrm{SL}(n,\mathbb{R})$. By exploiting these properties and extending the approach in [23], we establish the following result:

Theorem 11 *Let $P = \mathrm{DS}(\mathbf{X}, \Gamma_0)$ be an exact partial Dirichlet–Selberg domain centered at $\mathbf{X} \in \mathcal{X}_3$, defined with respect to a finite set $\Gamma_0 \subset \mathrm{SL}(3,\mathbb{R})$, with a proper canonical facet pairing. If, in addition, P has finite volume, then the quotient of P under the facet pairing is complete.*

1.4 Organization of the Paper

This paper is structured as follows. In Section 2, we introduce the key construction - **Busemann-Selberg functions** on $\mathcal{X}_n$, define generalized horospheres via these functions. In Section 3, we establish various properties of Busemann-Selberg functions, horospheres and hyperplanes, which are required for the proof of the main theorem. Section 4 presents the proof of Theorem 11, synthesizing earlier results. Finally, we give concrete examples in Section 5, constructing exact finitely-sided Dirichlet-Selberg domains in $\mathcal{X}_n$ that illustrate our main results.

2 Busemann-Selberg Function and Horoballs in $\mathcal{X}_n$

By replacing the hyperbolic distance with Selberg's invariant $\mathfrak{s}$ on the symmetric space $\mathcal{X}_n$, we extend the classical Busemann function and define the **Busemann-Selberg functions** in this section. We also discuss their level sets, or **horoballs**, in $\mathcal{X}_n$.

2.1 Busemann-Selberg Functions and Horoballs in $\mathcal{X}_n$

Let us first consider **rays**, or one-dimensional half-planes in $\mathcal{X}_n$ (which are not geodesic in general). Given $\mathbf{X} \in \mathcal{X}_{n,\mathrm{hyp}}$ and a Satake point $[\alpha] \in \partial_S \mathcal{X}_{n,\mathrm{proj}}$, we consider the ray $\mathbf{A}(t)$ starting at $\mathbf{X}$ and heading toward $[\alpha]$. In the projective model, this ray can be represented as

$$[\mathbf{A}(t)] = [\mathbf{X} + t\alpha], \quad t \in [0, \infty),$$

where α is any representative matrix of the corresponding Satake point.

Definition 21 *For $\mathbf{X} \in \mathcal{X}_{n,hyp}$ and any representative matrix α of a Satake point $[\alpha] \in \partial_S \mathcal{X}_{n,proj}$, let $\mathbf{A}(t) \subset \mathcal{X}_{n,hyp}$ be the unique lift of the ray $[\mathbf{X}+t\alpha]$ to the hypersurface model. Define the* ***Busemann-Selberg function (of type n, or maximal type)*** *$\mathfrak{b}_{\alpha,\mathbf{X}} : \mathcal{X}_{n,hyp} \to \mathbb{R}$ by*

$$\mathfrak{b}_{\alpha,\mathbf{X}}(\mathbf{Y}) = \lim_{t\to\infty} \log \frac{\mathfrak{s}(\mathbf{Y}, \mathbf{A}(t))}{\mathfrak{s}(\mathbf{X}, \mathbf{A}(t))}, \quad \forall \mathbf{Y} \in \mathcal{X}_n.$$

Remark 21 *The definition is independent of the choice of representative matrix for α. Indeed, the definition of $\mathfrak{s}$ implies the closed form*

$$\mathfrak{b}_{\alpha,\mathbf{X}}(\mathbf{Y}) = \log \frac{\mathrm{tr}(\mathbf{Y}^{-1}\alpha)}{\mathrm{tr}(\mathbf{X}^{-1}\alpha)}.$$

This is invariant under any nonzero scaling $\alpha \mapsto c\alpha$, ensuring that the function is well defined with respect to the point $[\alpha] \in \partial_S \mathcal{X}_{n,proj}$.

As discussed in the Introduction, the proof of the main theorem requires a further generalization of the Busemann-Selberg function. This is obtained by composing a maximal-type Busemann-Selberg function with the projection onto a Satake boundary component, as introduced below.

Definition 22 *Let $V \subset \mathbb{R}^n$ be a k-dimensional subspace, and let $\iota_V = (\mathbf{v}_1, \ldots, \mathbf{v}_k) \in \mathrm{Mat}_{n\times k}(\mathbb{R})$ be a matrix representing an orthonormal basis. The inverse of the embedding described in Proposition 11 gives a bijection $\pi_V : \partial_S(V) \subset \partial_S \mathcal{X}_n \to \mathcal{X}_{k,proj}$, where*

$$\pi_V([\alpha]) = [\iota_V^{\mathsf{T}}\, \alpha\, \iota_V].$$

This extends to a projection $\pi_V : \mathcal{X}_n \sqcup \mathrm{st}(\partial_S(V)) \to \mathcal{X}_k$ by the same formula.

While π_V depends on the choice of basis ι_V, any two such maps $\pi_{V,1}$ and $\pi_{V,2}$ differ only by post-composition with an element of $\mathrm{SO}(k)$ acting on $\mathcal{X}_k$.

The conjugation of π_V by the inversion map admits a continuous extension to the corresponding boundary component:

Proposition 21 *Define $\widetilde{\pi}_V : \mathcal{X}_n \to \mathcal{X}_k$ as the conjugation of π_V by the inversion map:*

$$\widetilde{\pi}_V(\mathbf{X}) = (\pi_V(\mathbf{X}^{-1}))^{-1}.$$

Then $\widetilde{\pi}_V$ continuously extends to a map from $\mathcal{X}_n \sqcup \overline{\partial_S(V)}$ to $\overline{\mathcal{X}_k}$.

Proof Let α be a representative matrix of a point in $\overline{\partial_S(V)}$. By the definition of $\partial_S(V)$, there exists a unique positive semi-definite matrix $\mathbf{Y} \in \mathrm{Mat}_{k\times k}(\mathbb{R})$ such that $\alpha = \iota_V \mathbf{Y} \iota_V^{\mathsf{T}}$. Take any $\mathbf{X} \in \mathcal{X}_n$ and consider the ray $[\mathbf{X} + t\iota_V \mathbf{Y} \iota_V^{\mathsf{T}}]$ for $t \in \mathbb{R}_+$ from $\mathbf{X}$ toward $[\alpha]$. This ray can also be expressed as $[\epsilon\mathbf{X} + \iota_V \mathbf{Y} \iota_V^{\mathsf{T}}]$, where $\epsilon = t^{-1}$. We claim that:

$$\lim_{\epsilon\to 0^+} (\iota_V^{\mathsf{T}}(\epsilon\mathbf{X} + \iota_V \mathbf{Y} \iota_V^{\mathsf{T}})^{-1}\iota_V)^{-1} = \mathbf{Y}.$$

Indeed, since ι_V consists of orthonormal column vectors, there exists a matrix $\iota_{V'} \in \mathrm{Mat}_{n\times(n-k)}(\mathbb{R})$ such that

$$\mathbf{U} := (\iota_V \mid \iota_{V'}) \in \mathrm{SO}(n).$$

We now represent $\mathbf{X}$ with respect to the basis $\mathbf{U}$:

$$\mathbf{U}^{\mathsf{T}}\mathbf{X}\mathbf{U} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^{\mathsf{T}} & \mathbf{C} \end{pmatrix},$$

where $\mathbf{A}$ is a $k \times k$ matrix. Note that $\iota_V^{\mathsf{T}}(\epsilon\mathbf{X} + \iota_V \mathbf{Y} \iota_V^{\mathsf{T}})^{-1}\iota_V$ is the $(1,1)$-block of the following matrix:

$$\mathbf{U}^{\mathsf{T}}(\epsilon\mathbf{X} + \iota_V \mathbf{Y} \iota_V^{\mathsf{T}})^{-1}\mathbf{U} = (\epsilon\mathbf{U}^{\mathsf{T}}\mathbf{X}\mathbf{U} + \mathbf{U}^{\mathsf{T}}\iota_V \mathbf{Y} \iota_V^{\mathsf{T}}\mathbf{U})^{-1} = \begin{pmatrix} \mathbf{Y} + \epsilon\mathbf{A} & \epsilon\mathbf{B} \\ \epsilon\mathbf{B}^{\mathsf{T}} & \epsilon\mathbf{C} \end{pmatrix}^{-1},$$

where the last equality follows from the orthogonality of $\mathbf{U}$, which implies $\iota_V^{\mathsf{T}}\mathbf{U} = (\mathbf{I}_k \mid \mathbf{O})$.

By the block inverse formula, the inverse of this $(1,1)$-block is:

$$(\iota_V^{\mathsf{T}}(\epsilon\mathbf{X}+\iota_V \mathbf{Y} \iota_V^{\mathsf{T}})^{-1}\iota_V)^{-1} = \mathbf{Y}+\epsilon\mathbf{A}-(\epsilon\mathbf{B})(\epsilon\mathbf{C})^{-1}\epsilon\mathbf{B}^{\mathsf{T}} = \mathbf{Y}+\epsilon(\mathbf{A}-\mathbf{B}\mathbf{C}^{-1}\mathbf{B}^{\mathsf{T}}),$$

which converges to $\mathbf{Y}$ as $\epsilon \to 0^+$. Thus, $\widetilde{\pi}_V$ extends continuously such that $\widetilde{\pi}_V([\alpha]) = [\mathbf{Y}]$ for every $\alpha = \iota_V \mathbf{Y} \iota_V^{\mathsf{T}}$ representing a point in $\overline{\partial_S(V)}$. □

For a subspace $V \subset \mathbb{R}^n$ of dimension k, we call the composition of $\widetilde{\pi}_V : \mathcal{X}_n \to \mathcal{X}_k$ with a maximal-type Busemann-Selberg function on $\mathcal{X}_k$ a **Busemann-Selberg function of type** k on $\mathcal{X}_n$.

Definition 23 *Let $\mathbf{X} \in \mathcal{X}_n$, and let $\Pi = \partial_S(V)$ be a boundary component of type k. Suppose that α lies on $\partial\Pi$, so that* $\mathrm{rank}(\alpha) < k$.

*The **type-k Busemann-Selberg function** based at $(\alpha, \partial_S(V))$ with reference point $\mathbf{X}$ is the function $\mathfrak{b}^{(k)}_{V;\alpha,\mathbf{X}} : \mathcal{X}_n \to \mathbb{R}$ defined by*

$$\mathfrak{b}^{(k)}_{V;\alpha,\mathbf{X}}(\mathbf{Y}) = \mathfrak{b}_{\widetilde{\pi}_V(\alpha),\widetilde{\pi}_V(\mathbf{X})}(\widetilde{\pi}_V(\mathbf{Y})).$$

For convenience, if the Satake boundary component $\partial_S(V)$ is denoted by Π, we also write π_Π and $\widetilde{\pi}_\Pi$ for the maps π_V and $\widetilde{\pi}_V$, respectively. We also denote the corresponding Busemann-Selberg function by $\mathfrak{b}^{(k)}_{\Pi;\alpha,\mathbf{X}}$.

Proposition 22 *If π_V is defined by a matrix $\iota_V \in \mathrm{Mat}_{n\times k}(\mathbb{R})$ representing an orthonormal basis of V, then the Busemann-Selberg function of type k takes the closed form*

$$\mathfrak{b}^{(k)}_{V;\alpha,\mathbf{X}}(\mathbf{Y}) = \log \frac{\mathrm{tr}(\mathbf{Y}^{-1}\alpha)}{\mathrm{tr}(\mathbf{X}^{-1}\alpha)} - \frac{1}{k}\log \frac{\det(\iota_V^{\mathsf{T}}\mathbf{Y}^{-1}\iota_V)}{\det(\iota_V^{\mathsf{T}}\mathbf{X}^{-1}\iota_V)}, \tag{1}$$

for $\mathbf{X}, \mathbf{Y} \in \mathcal{X}_{n,hyp}$ and any representative α of $[\alpha] \in \mathcal{X}_{n,proj}$.

Proof As we noted in the proof of Proposition 21, there is a positive semi-definite matrix $\mathbf{A} \in \mathrm{Mat}_{k\times k}(\mathbb{R})$ such that $\alpha = \iota_V \mathbf{A} \iota_V^{\mathsf{T}}$, and $\widetilde{\pi}_V([\alpha]) = [\mathbf{A}]$.

On the other hand, the projection $\widetilde{\pi}_V(\mathbf{X}) = [(\iota_V^{\mathsf{T}}\mathbf{X}^{-1}\iota_V)^{-1}]$ in the projective model, so in the hypersurface model,

$$\widetilde{\pi}_V(\mathbf{X}) = \frac{(\iota_V^{\mathsf{T}}\mathbf{X}^{-1}\iota_V)^{-1}}{\big(\det(\iota_V^{\mathsf{T}}\mathbf{X}^{-1}\iota_V)^{-1}\big)^{1/k}} = (\iota_V^{\mathsf{T}}\mathbf{X}^{-1}\iota_V)^{-1}\big(\det(\iota_V^{\mathsf{T}}\mathbf{X}^{-1}\iota_V)\big)^{1/k}.$$

As in the definition, $\mathfrak{b}^{(k)}_{V;\alpha,\mathbf{X}}(\mathbf{Y}) = \log \frac{\mathrm{tr}(\widetilde{\pi}_V(\mathbf{Y})^{-1}\widetilde{\pi}_V(\alpha))}{\mathrm{tr}(\widetilde{\pi}_V(\mathbf{X})^{-1}\widetilde{\pi}_V(\alpha))}$, where

$$\begin{aligned}\mathrm{tr}(\widetilde{\pi}_V(\mathbf{X})^{-1}\widetilde{\pi}_V(\alpha)) &= \frac{\mathrm{tr}(\iota_V^{\mathsf{T}}\mathbf{X}^{-1}\iota_V\mathbf{A})}{\big(\det(\iota_V^{\mathsf{T}}\mathbf{X}^{-1}\iota_V)\big)^{1/k}} \\ &= \frac{\mathrm{tr}(\mathbf{X}^{-1}\iota_V\mathbf{A}\iota_V^{\mathsf{T}})}{\big(\det(\iota_V^{\mathsf{T}}\mathbf{X}^{-1}\iota_V)\big)^{1/k}} = \frac{\mathrm{tr}(\mathbf{X}^{-1}\alpha)}{\big(\det(\iota_V^{\mathsf{T}}\mathbf{X}^{-1}\iota_V)\big)^{1/k}},\end{aligned}$$

and similarly

$$\mathrm{tr}(\widetilde{\pi}_V(\mathbf{Y})^{-1}\widetilde{\pi}_V(\alpha)) = \frac{\mathrm{tr}(\mathbf{Y}^{-1}\alpha)}{\big(\det(\iota_V^{\mathsf{T}}\mathbf{Y}^{-1}\iota_V)\big)^{1/k}},$$

leading to (1). □

Remark 22 *One can see from* (1) *that the definition of $\mathfrak{b}^{(k)}_{V;\alpha,\mathbf{X}}$ is independent of the choice of the matrix ι_V representing an orthonormal basis of V: a different choice takes the form $\iota_V g$ for certain $g \in \mathrm{SO}(k)$, while it holds that*

$$\det((\iota_{\Pi} g)^{\mathsf{T}}\mathbf{X}^{-1}(\iota_{\Pi} g)) = \det(g)^2 \det(\iota_{\Pi}^{\mathsf{T}}\mathbf{X}^{-1}\iota_{\Pi}) = \det(\iota_{\Pi}^{\mathsf{T}}\mathbf{X}^{-1}\iota_{\Pi}).$$

Let us consider the effect of the action of $g \in \mathrm{SL}(n,\mathbb{R})$ on Busemann-Selberg functions. If ι_V is a matrix whose columns are orthonormal vectors for $\Pi = \partial_S(V)$ (of type k), then $g^{\mathsf{T}}\iota_V$ is a matrix for $g.\Pi$. There is a matrix $h \in \mathrm{SL}(k,\mathbb{R})$ such that $g^{\mathsf{T}}\iota_V h$ consists of orthonormal column vectors for $g.\Pi$. The relations

$$\mathrm{tr}((g^{\mathsf{T}}\mathbf{Y}g)^{-1}(g^{\mathsf{T}}\alpha g)) = \mathrm{tr}(g^{-1}\mathbf{Y}^{-1}\alpha g) = \mathrm{tr}(\mathbf{Y}^{-1}\alpha),$$

and

$$\det(h^{\mathsf{T}}\iota_V^{\mathsf{T}} g (g^{\mathsf{T}}\mathbf{Y}g)^{-1} g^{\mathsf{T}}\iota_V h) = \det(h^{\mathsf{T}}\iota_V^{\mathsf{T}}\mathbf{Y}^{-1}\iota_V h) = \det(\iota_V^{\mathsf{T}}\mathbf{Y}^{-1}\iota_V),$$

implies the following preservation property.

Proposition 23 *For any $\alpha \in \partial_S \mathcal{X}_n$, $\mathbf{X}$, $\mathbf{Y} \in \mathcal{X}_n$, and $g \in \mathrm{SL}(n,\mathbb{R})$,*

$$\mathfrak{b}_{g.\alpha,g.\mathbf{X}}(g.\mathbf{Y}) = \mathfrak{b}_{\alpha,\mathbf{X}}(\mathbf{Y}),$$

and for any type k boundary component Π with $\alpha \in \partial\Pi$,

$$\mathfrak{b}^{(k)}_{g.\Pi;g.\alpha,g.\mathbf{X}}(g.\mathbf{Y}) = \mathfrak{b}^{(k)}_{\Pi;\alpha,\mathbf{X}}(\mathbf{Y}).$$

We shall refer to the sublevel sets of the Busemann-Selberg functions as **horoballs**, and their level sets as **horospheres**.

Definition 24 *Let $\alpha \in \partial_S \mathcal{X}_n$ be a Satake boundary point, and let $\mathbf{X} \in \mathcal{X}_n$ be a reference point. For each $r \in \mathbb{R}$, the **closed horoball** based at α with parameter r is*

$$\mathrm{B}_{\mathbf{X}}(\alpha, r) = \{\mathbf{Y} \in \mathcal{X}_n \mid \mathfrak{b}_{\alpha,\mathbf{X}}(\mathbf{Y}) \leq r\}.$$

*Replacing "$\leq$" by "$<$" yields the corresponding **open horoball**.*

*The **horosphere** at level r is the level set*

$$\Sigma_{\mathbf{X}}(\alpha, r) = \{\mathbf{Y} \in \mathcal{X}_n \mid \mathfrak{b}_{\alpha,\mathbf{X}}(\mathbf{Y}) = r\}.$$

This construction generalizes to non-maximal type settings:

Definition 25 *Let Π be a boundary component of type k, let α lie on $\partial\Pi$, and let $\mathbf{X} \in \mathcal{X}_n$ be a reference point. For each $r \in \mathbb{R}$, define the **type-k horoball** based at (α, Π) with parameter r by*

$$\mathrm{B}^{(k)}_{\Pi;\mathbf{X}}(\alpha, r) = \{\mathbf{Y} \in \mathcal{X}_n \mid \mathfrak{b}^{(k)}_{\Pi;\alpha,\mathbf{X}}(\mathbf{Y}) \leq r\}.$$

*Similarly, the **type-k horosphere** at level r is defined as*

$$\Sigma^{(k)}_{\Pi;\mathbf{X}}(\alpha, r) = \{\mathbf{Y} \in \mathcal{X}_n \mid \mathfrak{b}^{(k)}_{\Pi;\alpha,\mathbf{X}}(\mathbf{Y}) = r\}.$$

As before, when $\Pi = \partial_S(V)$, we also use the notations $\mathrm{B}^{(k)}_{V;\mathbf{X}}(\alpha, r)$ and $\Sigma^{(k)}_{V;\mathbf{X}}(\alpha, r)$.

Example 21 *Take the reference point $\mathbf{X} = \mathbf{I}_3$ in the symmetric space $\mathcal{X}_3$. For $[\alpha] \in \partial_S \mathcal{X}_3$ represented by $\alpha = \mathrm{diag}(1,0,0)$, the maximal-type Busemann-Selberg function $\mathfrak{b}_{\alpha,\mathbf{X}}$ is computed as*

$$\mathfrak{b}_{\alpha,\mathbf{X}}(\mathbf{Y}) = \log(y_{22}y_{33} - y_{23}^2),$$

for $\mathbf{Y} = (y_{ij}) \in \mathcal{X}_{3,hyp}$.

For a different choice of Satake point, $\beta = \mathrm{diag}(1,1,0)$, the corresponding Busemann-Selberg function is computed as

$$\mathfrak{b}_{\beta,\mathbf{X}}(\mathbf{Y}) = \log \frac{y_{22}y_{33} - y_{23}^2 + y_{11}y_{33} - y_{13}^2}{2}.$$

For $V = \mathrm{span}(\mathbf{e}_1, \mathbf{e}_2) \subset \mathbb{R}^3$, the type-2 Busemann-Selberg function $\mathfrak{b}^{(2)}_{V;\alpha,\mathbf{X}}$ takes the form

$$\mathfrak{b}^{(2)}_{V;\alpha,\mathbf{X}}(\mathbf{Y}) = \log(y_{22}y_{33} - y_{23}^2) - \frac{1}{2}\log y_{33}.$$

Similarly, for $W = \mathrm{span}(\mathbf{e}_1, \mathbf{e}_3)$,

$$\mathfrak{b}^{(2)}_{W;\alpha,\mathbf{X}}(\mathbf{Y}) = \log(y_{22}y_{33} - y_{23}^2) - \frac{1}{2}\log y_{22}.$$

While visualizing the horospheres corresponding to these Busemann-Selberg functions in a five-dimensional space is impractical, we consider their intersections with the model maximal flat $F_{mod} \subset \mathcal{X}_3$*, which is the* 2*-plane of positive definite diagonal matrices.*

Let us display F_{mod} *in two ways. When realizing* F_{mod} *isometrically as a Euclidean* 2*-space, the horoballs intersect* F_{mod} *in convex regions. In particular, the horoballs based at* α *intersect it as half-planes.*

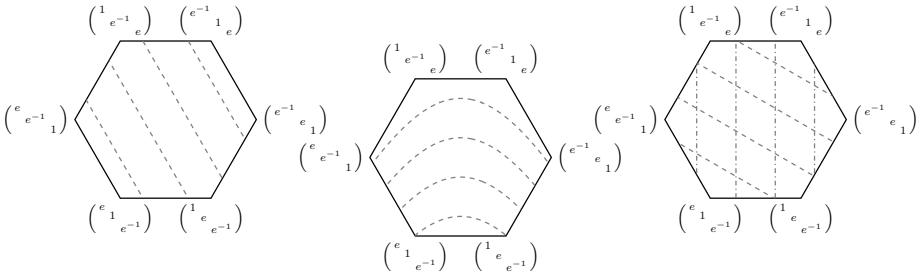


Fig. 1 Intersections of horospheres with F_{mod}, restricted to a hexagon. Left: horospheres $\Sigma(\alpha, r)$ for varying r. Center: horospheres $\Sigma(\beta, r)$. Right: type-2 horospheres $\Sigma^{(2)}_V(\alpha, r)$ (dash-dot) and $\Sigma^{(2)}_W(\alpha, r)$ (dashed) superimposed.

When realizing F_{mod} *as the interior of the projective triangle spanned by* $\mathbf{e}_i \otimes \mathbf{e}_i$*,* $i = 1, 2, 3$*, the maximal-type horoballs intersect* F_{mod} *in regions that are not projectively convex.*

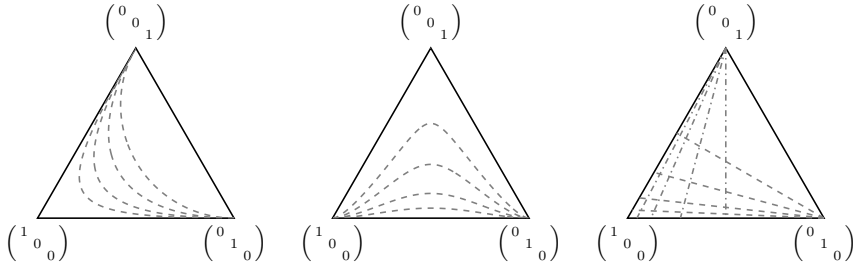


Fig. 2 Intersections of same families of horospheres with F_{mod}; the latter is expressed as a projective triangle.

2.2 Lipschitz Continuity for Busemann-Selberg Functions

Busemann-Selberg functions can be expressed in terms of the classical Busemann functions $\mathrm{b}_{\xi,\mathbf{X}}$. In particular, when α is a rank-one Satake point, a (type-k) Busemann-Selberg function decomposes as a linear combination of the corresponding Busemann functions.

Let $\mathbf{A} = \mathrm{diag}(a_1, \ldots, a_n)$, with $a_i \geq a_{i+1}$, $\sum_{i=1}^n a_i = 0$, and

$$\|\mathbf{A}\|^2 = 2n \sum_{i=1}^n a_i^2 = 1,$$

so that the parametrized geodesic ray $\gamma : \mathbb{R}_{\geq 0} \to \mathcal{X}_n$, $\gamma(t) = \exp(t\mathbf{A})$, is unit-speed. Let $\xi = \gamma(\infty)$, which is a point on the visual boundary of $\mathcal{X}_n$. The Busemann function

$$\mathrm{b}_{\xi,\mathbf{I}_n}(\mathbf{X}) = \lim_{t\to\infty} \big(d(\mathbf{X}, \gamma(t)) - t\big),\ \mathbf{X} \in \mathcal{X}_{n,\mathrm{hyp}},$$

is explicitly given by [16] (Lemma 2-5), taking σ in Hattori's formula as the reversal permutation of $\{1, 2, \ldots, n\}$:

$$\mathrm{b}_{\xi,\mathbf{I}_n}(\mathbf{X}) = n \sum_{k=1}^{n-1} (a_k - a_{k+1}) \log \mathbf{X}_{[k]}^{-1},$$

where $\mathbf{X}_{[k]}^{-1}$ denotes the k-th upper-left minor of $\mathbf{X}^{-1}$, which agrees with the $(n-k)$-th lower-right minor of $\mathbf{X}$ appearing in their formula.

For $k = 1, \ldots, n-1$, the one-step flag $\{\mathbf{0}\} \subset \mathrm{span}(\mathbf{e}_1, \ldots, \mathbf{e}_k) \subset \mathbb{R}^n$ is a vertex on the ideal boundary, which corresponds to the following diagonal matrix $\mathbf{A}$ with only two distinct eigenvalues:

$$\mathbf{A}_k = \mathrm{diag}\big(\sqrt{\frac{(n-k)}{2kn^2}}\mathbf{I}_k, -\sqrt{\frac{k}{2(n-k)n^2}}\mathbf{I}_{n-k}\big).$$

The Busemann function for $\xi_k = \gamma_k(\infty)$, where $\gamma_k(t) = \exp(t\mathbf{A}_k)$, is

$$\mathrm{b}_{\xi_k,\mathbf{I}_n}(\mathbf{X}) = \frac{n}{\sqrt{2k(n-k)}} \log \mathbf{X}_{[k]}^{-1}$$

thus

$$\mathrm{b}_{\xi_k,\mathbf{X}}(\mathbf{Y}) = \frac{n}{\sqrt{2k(n-k)}} \log \frac{\mathbf{Y}_{[k]}^{-1}}{\mathbf{X}_{[k]}^{-1}}.$$

For $\alpha = \mathbf{e}_1 \otimes \mathbf{e}_1 \in \partial_S \mathcal{X}_n$ and $\mathit{\Pi} = \partial_S(\mathbf{e}_1, \ldots, \mathbf{e}_k)$, equation (1) can be expressed as

$$\mathfrak{b}_{\alpha,\mathbf{X}} = \frac{\sqrt{2(n-1)}}{n} \mathrm{b}_{\xi_1,\mathbf{X}}(\mathbf{Y})$$

and

$$\mathfrak{b}^{(k)}_{\mathit{\Pi};\alpha,\mathbf{X}} = \frac{\sqrt{2(n-1)}}{n} \mathrm{b}_{\xi_1,\mathbf{X}}(\mathbf{Y}) - \frac{\sqrt{2(n-k)}}{n\sqrt{k}} \mathrm{b}_{\xi_k,\mathbf{X}}(\mathbf{Y}).$$

The $\mathrm{SL}(n,\mathbb{R})$-equivariance extends this result to type-one Satake points α and boundary components $\mathit{\Pi}$ corresponding to general one-step flags:

Proposition 24 *Let $\Pi = \partial_S(V)$ be a boundary component of type $k \geq 2$, and let $\alpha = \mathbf{v} \otimes \mathbf{v}$ be a Satake point on $\partial\Pi$ (so $\mathbf{v} \in V$). Denote by $\xi_{\mathbf{v}}$ and ξ_V the corresponding vertices in the visual boundary $\partial_\infty \mathcal{X}_n$. Then for all $\mathbf{X}, \mathbf{Y} \in \mathcal{X}_n$:*

$$\mathfrak{b}_{\alpha,\mathbf{X}}(\mathbf{Y}) = \frac{\sqrt{2(n-1)}}{n} \mathrm{b}_{\xi_{\mathbf{v}},\mathbf{X}}(\mathbf{Y}),$$
$$\mathfrak{b}^{(k)}_{\Pi;\alpha,\mathbf{X}}(\mathbf{Y}) = \frac{\sqrt{2(n-1)}}{n} \mathrm{b}_{\xi_{\mathbf{v}},\mathbf{X}}(\mathbf{Y}) - \frac{\sqrt{2(n-k)}}{n\sqrt{k}} \mathrm{b}_{\xi_V,\mathbf{X}}(\mathbf{Y}).$$

Since higher rank α decompose as sums of rank-one matrices, every Busemann-Selberg function (of any type) can be written in terms of the original Busemann functions $\mathrm{b}_{\xi,\mathbf{X}}$.

The main result in this subsection is the following Lipschitz continuity for Busemann-Selberg functions.

Proposition 25 *Let Π and α be as above, and let $\mathbf{X}, \mathbf{Y}_1, \mathbf{Y}_2 \in \mathcal{X}_n$. Then*

$$|\mathfrak{b}^{(k)}_{\Pi;\alpha,\mathbf{X}}(\mathbf{Y}_1) - \mathfrak{b}^{(k)}_{\Pi;\alpha,\mathbf{X}}(\mathbf{Y}_2)| \leq \sqrt{\frac{2(k-1)}{k\ n}} d(\mathbf{Y}_1, \mathbf{Y}_2).$$

Lemma 21 *The projection $\pi : \mathcal{X}_n \to \mathcal{X}_k$ from Definition 22 is $\sqrt{\frac{k}{n}}$-Lipschitz, i.e.,*

$$d(\pi(\mathbf{Y}_1), \pi(\mathbf{Y}_2)) \leq \sqrt{\frac{k}{n}} d(\mathbf{Y}_1, \mathbf{Y}_2).$$

Proof First, we prove the lemma in the case $\mathbf{Y}_1 = \mathbf{I}_n$. Since π is given by conjugation with an orthonormal-column matrix ι, we have $\pi(\mathbf{Y}_1) = \iota^\mathsf{T}\iota = \mathbf{I}_k$. Hence it suffices to prove

$$d\left(\mathbf{I}_k, \frac{\iota^\mathsf{T}\mathbf{Y}\iota}{\det(\iota^\mathsf{T}\mathbf{Y}\iota)^{1/k}}\right) \leq \sqrt{\frac{k}{n}} d(\mathbf{I}_n, \mathbf{Y}),$$

where $\mathbf{Y} = \mathbf{Y}_2$. Let $\lambda_1 \geq \cdots \geq \lambda_n$ be the eigenvalues of $\mathbf{Y}$, and let $\mu_1 \geq \cdots \geq \mu_k$ be those of $\iota^\mathsf{T}\mathbf{Y}\iota$. By the Poincaré separation theorem,

$$\lambda_i \geq \mu_i \geq \lambda_{i+n-k},\ i = 1, \ldots, k.$$

Then, using Lemma C1 in the Appendix and the fact that $\sum_i \log \lambda_i = 0$, we obtain

$$d^2(\mathbf{I}_k, \iota^\mathsf{T}\mathbf{Y}\iota / \det(\iota^\mathsf{T}\mathbf{Y}\iota)^{1/k})$$
$$= (2k) \sum_{i=1}^{k} (\log \mu_i - \overline{\log \mu})^2 \leq (2k) \sum_{i=1}^{n} (\log \lambda_i)^2 = \frac{k}{n} d^2(\mathbf{I}_n, \mathbf{Y}),$$

which gives the desired bound.

For general $\mathbf{Y}_1 \in \mathcal{X}_{n,\mathrm{hyp}}$, let $g = \mathbf{Y}_1^{-1/2} \in \mathrm{SL}(n, \mathbb{R})$, so $g^\mathsf{T}\mathbf{Y}_1 g = \mathbf{I}_n$. For any orthonormal-column $\iota \in \mathrm{Mat}_{n\times k}(\mathbb{R})$, polar decomposition yields another orthonormal-column matrix $\iota' \in \mathrm{Mat}_{n\times k}(\mathbb{R})$ and $h \in \mathrm{GL}^+(k, \mathbb{R})$, such that

$$g^{-1}\iota = \iota' h^{-1},\ \text{or}\ g\iota' = \iota h.$$

Using the conclusion for $\mathbf{Y}_1 = \mathbf{I}_n$:

$$\begin{aligned}
&\sqrt{\frac{k}{n}} d(\mathbf{Y}_1, \mathbf{Y}_2) = \sqrt{\frac{k}{n}} d(\mathbf{I}_n, g^\mathsf{T} \mathbf{Y}_2 g) \\
&\geq d(\mathbf{I}_k, [(g\iota')^\mathsf{T} \mathbf{Y}_2 (g\iota')]) = d([(g\iota')^\mathsf{T} \mathbf{Y}_1 (g\iota')], [(g\iota')^\mathsf{T} \mathbf{Y}_2 (g\iota')]) \\
&= d([(\iota h)^\mathsf{T} \mathbf{Y}_1 (\iota h)], [(\iota h)^\mathsf{T} \mathbf{Y}_2 (\iota h)]) = d([h^\mathsf{T} (\iota^\mathsf{T} \mathbf{Y}_1 \iota) h], [h^\mathsf{T} (\iota^\mathsf{T} \mathbf{Y}_2 \iota) h]) \\
&= d([\iota^\mathsf{T} \mathbf{Y}_1 \iota], [\iota^\mathsf{T} \mathbf{Y}_2 \iota]) = d(\pi(\mathbf{Y}_1), \pi(\mathbf{Y}_2)).
\end{aligned}$$

Here the last line holds because the action of $h/\det(h)^{1/k} \in \mathrm{SL}(k, \mathbb{R})$ preserves the distance in $\mathcal{X}_k$. □

Proof (Proposition 25) For $k = n$, the Lipschitz continuity for $\mathfrak{b}_{\alpha,\mathbf{X}}$ with $\mathrm{rank}(\alpha) = 1$ follows from Proposition 24. Any higher rank α decomposes into rank-1 summands, so the result extends by linearity.

For $k < n$, one reduces to the maximal-type case in $\mathcal{X}_k$:

$$\begin{aligned}
&|\mathfrak{b}^{(k)}_{\Pi;\alpha,\mathbf{X}}(\mathbf{Y}_1) - \mathfrak{b}^{(k)}_{\Pi;\alpha,\mathbf{X}}(\mathbf{Y}_2)| \\
&= |\mathfrak{b}_{\widetilde{\pi}(\alpha),\widetilde{\pi}(\mathbf{X})}(\widetilde{\pi}(\mathbf{Y}_1)) - \mathfrak{b}_{\widetilde{\pi}(\alpha),\widetilde{\pi}(\mathbf{X})}(\widetilde{\pi}(\mathbf{Y}_2))| \\
&\leq \tfrac{\sqrt{2(k-1)}}{k} d(\widetilde{\pi}(\mathbf{Y}_1), \widetilde{\pi}(\mathbf{Y}_2)).
\end{aligned}$$

Since $d(\mathbf{Y}_1, \mathbf{Y}_2) = d(\mathbf{Y}_1^{-1}, \mathbf{Y}_2^{-1})$ and by Lemma 21,

$$\begin{aligned}
&d(\widetilde{\pi}(\mathbf{Y}_1), \widetilde{\pi}(\mathbf{Y}_2)) = d(\pi(\mathbf{Y}_1^{-1})^{-1}, \pi(\mathbf{Y}_2^{-1})^{-1}) = d(\pi(\mathbf{Y}_1^{-1}), \pi(\mathbf{Y}_2^{-1})) \\
&\leq \sqrt{\frac{k}{n}} d(\mathbf{Y}_1^{-1}, \mathbf{Y}_2^{-1}) = \sqrt{\frac{k}{n}} d(\mathbf{Y}_1, \mathbf{Y}_2),
\end{aligned}$$

the proposition follows. □

2.3 Asymptotic Behavior of Busemann-Selberg Functions

Busemann-Selberg functions exhibit asymptotic behavior analogous to, but more intricate than, that of the classical hyperbolic Busemann functions. In this subsection, we describe this generalized phenomenon for points asymptotic to each Satake boundary component of $\mathcal{X}_n$.

Recall that the ray from $\mathbf{Y} \in \mathcal{X}_{n,\mathrm{hyp}}$ to $\beta \in \partial_S \mathcal{X}_n$ is expressed in the projective model as

$$[\mathbf{Y} + t\beta],\ t \in \mathbb{R}_+,\ \text{or } [\beta + \epsilon \mathbf{Y}],\ \epsilon = t^{-1} \in \mathbb{R}_+.$$

Therefore, to study the behavior of Busemann-Selberg functions along rays asymptotic to β, we consider their values at $\beta + \epsilon\mathbf{Y}$ as $\epsilon \to 0^+$. The first two lemmas directly generalize the asymptotic behavior of hyperbolic Busemann functions.

Lemma 22 *Let $\Pi \subset \overline{\mathcal{X}_n}$ be a boundary component of type k, pick $\alpha \in \partial\Pi$, and fix $\mathbf{X} \in \mathcal{X}_n$. Suppose $\beta \in \partial_S\mathcal{X}_n$ satisfies*

$$\alpha \notin \overline{\operatorname{supp}(\beta)} \text{ and } \overline{\Pi} \cap \overline{\operatorname{supp}(\beta)} \neq \varnothing.$$

Then for any $\mathbf{Y} \in \mathcal{X}_n$,

$$\lim_{\epsilon \to 0^+} \mathfrak{b}^{(k)}_{\Pi;\alpha,\mathbf{X}}(\beta + \epsilon\mathbf{Y}) = \infty.$$

Proof After an $\mathrm{SL}(n,\mathbb{R})$-action (Proposition 23), we may assume

$$\Pi = \partial_S(\mathbf{e}_1, \dots, \mathbf{e}_k),\ \alpha \in \partial_S(\mathbf{e}_i,\ i \in \mathsf{A}),\ \text{and } \beta \in \partial_S(\mathbf{e}_j,\ j \in \mathsf{B}),$$

where $\mathsf{A}, \mathsf{B} \subset \{1, \dots, n\}$. The conditions on α, β, and Π imply that $\mathsf{A} \setminus \mathsf{B} \neq \varnothing$ and $\mathsf{B} \cap \{1, \dots, k\} \neq \varnothing$.

Write

$$(\beta + \epsilon\mathbf{Y})^{-1} = \mathbf{M}_{-1}\epsilon^{-1} + \mathbf{M}_0 + O(\epsilon).$$

The coefficient matrix $\mathbf{M}_{-1}$ of the leading term is positive semi-definite, while its restriction to $\operatorname{span}(\mathbf{e}_i,\ i \in \mathsf{A} \setminus \mathsf{B})$ is positive definite. Hence

$$\operatorname{tr}((\beta + \epsilon\mathbf{Y})^{-1}\alpha) = O(\epsilon^{-1})$$

with a positive coefficient.

On the other hand, let $\iota_\Pi = (\mathbf{e}_1, \dots, \mathbf{e}_k)$. Since $\mathsf{B} \cap \{1, \dots, k\} \neq \varnothing$, the principal $k \times k$ minor of $\mathbf{M}_{-1}$ has at least one zero row and column. Therefore

$$\det(\iota_\Pi^{\mathsf{T}}(\beta + \epsilon\mathbf{Y})^{-1}\iota_\Pi) = o(\epsilon^{-k}).$$

Putting these estimates together,

$$\mathfrak{b}^{(k)}_{\Pi;\alpha,\mathbf{X}}(\beta + \epsilon\mathbf{Y}) \propto \log \frac{\operatorname{tr}((\beta + \epsilon\mathbf{Y})^{-1}\alpha)}{\det(\iota_\Pi^{\mathsf{T}}(\beta + \epsilon\mathbf{Y})^{-1}\iota_\Pi)^{1/k}} \xrightarrow{\epsilon \to 0^+} \infty,$$

as claimed. □

Lemma 23 *Let $\Pi \subset \overline{\mathcal{X}_n}$ be a boundary component of type k, choose $\alpha \in \partial\Pi$, and fix $\mathbf{X} \in \mathcal{X}_n$. Suppose $\beta \in \partial_S\mathcal{X}_n$ satisfies*

$$\alpha \in \overline{\operatorname{supp}(\beta)} \text{ and } \Pi \not\leq \operatorname{supp}(\beta).$$

Then for any $\mathbf{Y} \in \mathcal{X}_n$,

$$\lim_{\epsilon \to 0^+} \mathfrak{b}^{(k)}_{\Pi;\alpha,\mathbf{X}}(\beta + \epsilon\mathbf{Y}) = -\infty.$$

Proof After an $\mathrm{SL}(n,\mathbb{R})$-action, assume

$$\mathit{\Pi} = \partial_S(\mathbf{e}_1,\dots,\mathbf{e}_k),\ \alpha \in \partial_S(\mathbf{e}_i,\ i \in \mathsf{A}),\ \text{and}\ \beta \in \partial_S(\mathbf{e}_j,\ j \in \mathsf{B}),$$

where $\mathsf{A},\mathsf{B} \subset \{1,\dots,n\}$. The conditions on α, β, and $\mathit{\Pi}$ imply that $\mathsf{A} \subset \mathsf{B}$ and $\{1,\dots,k\} \setminus \mathsf{B} \neq \varnothing$.

Writing

$$(\beta + \epsilon\mathbf{Y})^{-1} = \mathbf{M}_{-1}\epsilon^{-1} + \mathbf{M}_0 + O(\epsilon),$$

one sees that $\mathbf{M}_{-1}$ vanishes on the columns indexed by B. Since $\mathsf{A} \subset \mathsf{B}$, we have

$$\mathrm{tr}((\beta + \epsilon\mathbf{Y})^{-1}\alpha) = O(1),$$

with a positive ϵ^0 coefficient.

On the other hand, let $\iota_{\mathit{\Pi}} = (\mathbf{e}_1,\dots,\mathbf{e}_k)$. Because $\{1,\dots,k\} \setminus \mathsf{B} \neq \varnothing$, the principal $k \times k$ minor of $\mathbf{M}_{-1}$ is nonzero and positive semi-definite, and this minor of $\mathbf{M}_0$ is positive definite. Thus

$$\det(\iota_{\mathit{\Pi}}^{\mathsf{T}}(\beta + \epsilon\mathbf{Y})^{-1}\iota_{\mathit{\Pi}}) = O(\epsilon^{-\delta}),$$

for some $\delta > 0$ depending on the cardinality of $\{1,\dots,k\} \setminus \mathsf{B}$. Therefore

$$\mathfrak{b}^{(k)}_{\mathit{\Pi};\alpha,\mathbf{X}}(\beta + \epsilon\mathbf{Y}) \propto \log \frac{\mathrm{tr}((\beta + \epsilon\mathbf{Y})^{-1}\alpha)}{\det(\iota_{\mathit{\Pi}}^{\mathsf{T}}(\beta + \epsilon\mathbf{Y})^{-1}\iota_{\mathit{\Pi}})^{1/k}} \xrightarrow{\epsilon\to 0^+} -\infty,$$

as desired. □

Two further asymptotic phenomena reveal the rich nature of Busemann-Selberg functions. To continue, notice that in the proof of Proposition 21, the formula

$$\widetilde{\pi}_V([\iota_V\mathbf{Y}\iota_V^{\mathsf{T}}]) = [\mathbf{Y}]$$

implies that $\widetilde{\pi}_V$ identifies the boundary components of $\partial_S(V)$ with the boundary components of $\mathcal{X}_k$. Therefore, if $\mathit{\Pi} < \partial_S(V)$, the notion of the boundary component $\widetilde{\pi}_V(\mathit{\Pi})$ of $\mathcal{X}_k$ is unambiguous.

Lemma 24 *Let $\mathit{\Pi} \subset \overline{\mathcal{X}_n}$ be a boundary component of type k, $\alpha \in \partial\mathit{\Pi}$, $\mathbf{X} \in \mathcal{X}_n$, and $\beta \in \partial_S\mathcal{X}_n$ satisfying*

$$\alpha \in \overline{\mathrm{supp}(\beta)}\ \textit{and}\ \mathit{\Pi} \leq \mathrm{supp}(\beta).$$

Then $\mathfrak{b}^{(k)}_{\mathit{\Pi};\alpha,\mathbf{X}}$ converges to a finite number at the point asymptotic to β. In particular, let $\mathit{\Xi} = \mathrm{supp}(\beta)$ of type $l \geq k$, and let

$$\widetilde{\pi} : \mathcal{X}_n \sqcup \overline{\mathit{\Xi}} \to \overline{\mathcal{X}_l}$$

be the projection described in Proposition 21. Then for each $\mathbf{Y} \in \mathcal{X}_n$,

$$\lim_{\epsilon\to 0^+} \mathfrak{b}^{(k)}_{\mathit{\Pi};\alpha,\mathbf{X}}(\beta + \epsilon\mathbf{Y}) = \mathfrak{b}^{(k)}_{\widetilde{\pi}(\mathit{\Pi});\widetilde{\pi}(\alpha),\widetilde{\pi}(\mathbf{X})}(\widetilde{\pi}(\beta)).$$

If $\mathit{\Pi} = \mathit{\Xi}$, the limit is a Busemann-Selberg function of maximal type.

Proof Denote by

$$\widetilde{\pi}_1 : \widetilde{\pi}(\mathcal{X}_n \sqcup \overline{\Pi}) = \mathcal{X}_l \sqcup \overline{\widetilde{\pi}(\Pi)} \to \overline{\mathcal{X}_k}$$

the canonical projection, so that $\widetilde{\pi}_1 \circ \widetilde{\pi}$ projects $\mathcal{X}_n \sqcup \overline{\Pi}$ to $\overline{\mathcal{X}_k}$. By the continuity of $\widetilde{\pi}_1$ and $\widetilde{\pi}$ on their domains, and since $\widetilde{\pi}(\beta)$ is invertible,

$$\begin{aligned}\lim_{\epsilon\to 0^+} \mathfrak{b}^{(k)}_{\Pi;\alpha,\mathbf{X}}(\beta+\epsilon\mathbf{Y}) &= \lim_{\epsilon\to 0^+} \log \frac{\mathrm{tr}(\widetilde{\pi}_1(\widetilde{\pi}(\beta+\epsilon\mathbf{Y}))^{-1}\widetilde{\pi}_1(\widetilde{\pi}(\alpha)))}{\mathrm{tr}(\widetilde{\pi}_1(\widetilde{\pi}(\mathbf{X}))^{-1}\widetilde{\pi}_1(\widetilde{\pi}(\alpha)))}\\ &= \log \frac{\mathrm{tr}(\widetilde{\pi}_1(\widetilde{\pi}(\beta))^{-1}\widetilde{\pi}_1(\widetilde{\pi}(\alpha)))}{\mathrm{tr}(\widetilde{\pi}_1(\widetilde{\pi}(\mathbf{X}))^{-1}\widetilde{\pi}_1(\widetilde{\pi}(\alpha)))} = \mathfrak{b}^{(k)}_{\widetilde{\pi}(\Pi);\widetilde{\pi}(\alpha),\widetilde{\pi}(\mathbf{X})}(\widetilde{\pi}(\beta)).\end{aligned}$$

□

Example 22 *Using the same notation as in Example 21:*

$$\mathbf{X} = \mathbf{I}_3 \in \mathcal{X}_3,\ \alpha = \mathrm{diag}(1,0,0) \in \partial_S\mathcal{X}_3,\ \textit{and } V = \mathrm{span}(\mathbf{e}_1,\mathbf{e}_2) \subset \mathbb{R}^3.$$

For $\mathbf{Y} = (y_{ij})_{i,j=1}^3 \in \mathcal{X}_{3,proj}$,

$$\mathfrak{b}^{(2)}_{V;\alpha,\mathbf{X}}(\mathbf{Y}) = \log \frac{y_{22}y_{33} - y_{23}^2}{\sqrt{y_{33}\det(\mathbf{Y})}}.$$

Therefore, for any $\mathbf{Y} \in \mathcal{X}_3$, *and*

$$\beta = \begin{pmatrix} \beta_0 & \mathbf{0} \\ \mathbf{0} & 0 \end{pmatrix} = \begin{pmatrix} b_{11} & b_{12} & 0 \\ b_{12} & b_{22} & 0 \\ 0 & 0 & 0 \end{pmatrix} \in \partial_S(V),$$

we have

$$\lim_{\epsilon\to 0^+} \mathfrak{b}^{(2)}_{V;\alpha,\mathbf{X}}(\beta+\epsilon\mathbf{Y}) = \lim_{\epsilon\to 0^+} \log \frac{\epsilon y_{33}(b_{22}+\epsilon y_{22}) - \epsilon^2 y_{23}^2}{\sqrt{\epsilon y_{33}(\epsilon y_{33}\det(\beta_0) + O(\epsilon^2))}} = \log \frac{b_{22}}{\sqrt{\det(\beta_0)}}.$$

This is the Busemann function of $[\beta_0] \in \mathcal{X}_{2,proj} = \mathbb{H}^2$ *based at* $\alpha_0 = \mathrm{diag}(1,0) \in \partial_\infty\mathbb{H}^2$.

Lemma 25 *Let* $\Pi \subset \overline{\mathcal{X}_n}$ *be a boundary component of type* k, $\alpha \in \partial\Pi$, $\mathbf{X} \in \mathcal{X}_n$, *and* $\beta \in \partial_S\mathcal{X}_n$ *satisfying*

$$\alpha \notin \overline{\mathrm{supp}(\beta)} \textit{ and } \overline{\Pi} \cap \overline{\mathrm{supp}(\beta)} = \varnothing.$$

Then the limit of $\mathfrak{b}^{(k)}_{\Pi;\alpha,\mathbf{X}}$ *as the point asymptotic to* β *depends on the path.*

Proof Let Ξ be a boundary component of type l, with $\Xi \geq \Pi$, and maximal among those satisfying $\overline{\mathrm{supp}(\beta)} \cap \overline{\Xi} = \varnothing$ (so $\mathrm{rank}(\beta) = n - l$). Denote by

$$\widetilde{\pi} : \mathcal{X}_n \sqcup \overline{\Xi} \to \overline{\mathcal{X}_l}$$

the projection described in Proposition 21. Up to an $\mathrm{SL}(n,\mathbb{R})$-action,

$$\Xi = \partial_S(\mathbf{e}_1,\ldots,\mathbf{e}_l),\ \text{and } \beta = \mathrm{diag}(\mathbf{O},\beta_3),$$

where $\beta_3 \in \mathrm{GL}(n-l,\mathbb{R})$.

For every $\mathbf{Y} \in \mathcal{X}_n$, we aim to show that

$$\lim_{\epsilon \to 0^+} \mathfrak{b}^{(k)}_{\Pi;\alpha,\mathbf{X}}(\beta + \epsilon\mathbf{Y}) = \mathfrak{b}^{(k)}_{\widetilde{\pi}(\Pi);\widetilde{\pi}(\alpha),\widetilde{\pi}(\mathbf{X})}(\pi(\mathbf{Y})).$$

Let

$$\mathbf{Y} = \begin{pmatrix} \mathbf{Y}_1 & \mathbf{Y}_2 \\ \mathbf{Y}_2^\intercal & \mathbf{Y}_3 \end{pmatrix},$$

where $\mathbf{Y}_1$ is an $l \times l$ block. The expression $\widetilde{\pi}([\beta + \epsilon\mathbf{Y}])$ is the inverse of the top-left $l \times l$ block of

$$(\beta + \epsilon\mathbf{Y})^{-1} = \begin{pmatrix} \epsilon\mathbf{Y}_1 & \epsilon\mathbf{Y}_2 \\ \epsilon\mathbf{Y}_2^\intercal & \beta_3 + \epsilon\mathbf{Y}_3 \end{pmatrix}^{-1}.$$

The block inverse formula shows that

$$\widetilde{\pi}([\beta + \epsilon\mathbf{Y}]) = [\epsilon\mathbf{Y}_1 - \epsilon^2\mathbf{Y}_2(\beta_3 + \epsilon\mathbf{Y}_3)^{-1}\mathbf{Y}_2^\intercal] = [\mathbf{Y}_1 - \epsilon\mathbf{Y}_2(\beta_3 + \epsilon\mathbf{Y}_3)^{-1}\mathbf{Y}_2^\intercal].$$

Since β_3 is invertible,

$$\lim_{\epsilon \to 0^+} \widetilde{\pi}([\beta + \epsilon\mathbf{Y}]) = \lim_{\epsilon \to 0^+} [\mathbf{Y}_1 - \epsilon\mathbf{Y}_2\beta_3^{-1}\mathbf{Y}_2^\intercal] = [\mathbf{Y}_1] = \pi(\mathbf{Y}).$$

The remainder of the argument follows exactly as in Lemma 24, by the continuity of the trace and determinant in the definition of $\mathfrak{b}^{(k)}_{\Pi;\alpha,\mathbf{X}}$. □

We summarize the asymptotic behavior of $\mathfrak{b}^{(k)}_{\Pi;\alpha,\mathbf{X}}$ at points asymptotic to β:

Conditions		Asymptotic behavior of $\mathfrak{b}^{(k)}_{\Pi;\alpha,\mathbf{X}}(\beta)$
$\alpha \in \overline{\mathrm{supp}(\beta)}$	$\Pi \leq \mathrm{supp}(\beta)$	Admits a finite limit
	$\Pi \not\leq \mathrm{supp}(\beta)$	Tends to $-\infty$
$\alpha \notin \overline{\mathrm{supp}(\beta)}$	$\overline{\Pi} \cap \overline{\mathrm{supp}(\beta)} = \varnothing$	Is path-dependent
	$\overline{\Pi} \cap \overline{\mathrm{supp}(\beta)} \neq \varnothing$	Tends to $+\infty$

3 Preliminary Lemmas for the Main Theorem

In the context of hyperbolic spaces, Proposition 14 on the completeness of Dirichlet domains is shown by combining the following facts: on the one hand, certain horospheres separate neighborhoods of ideal vertices from the remainder of the Dirichlet domain. On the other hand, Lemma 11 guarantees that the neighborhoods of ideal vertices glue into complete components, while the remainder itself has a complete quotient ([23], Theorem 11.1.2 and 11.1.4).

The situation is less satisfactory in higher rank settings. For one thing, an $\mathcal{X}_n$-polytope may contain higher-type and higher-dimensional Satake faces. In Subsection 3.1, we will show that the separation of neighborhoods of Satake faces by horospheres is also more intricate.

For another thing, the preservation of Busemann-Selberg functions by Satake face cycles is also less directly comparable to Lemma 11. We summarize the results in Subsection 3.2.

3.1 Tangency of Horospheres to the Satake Boundary

In hyperbolic space, any horosphere based at an ideal point $a \in \mathbb{H}^n$ meets the visual boundary only at a, and does so tangentially. A similar tangency phenomenon occurs for horospheres in $\mathcal{X}_n$, with additional cases to consider.

Throughout this subsection, we fix a reference point $\mathbf{X} \in \mathcal{X}_n$ for the Busemann-Selberg functions and horoballs. We omit $\mathbf{X}$ from the notation when the meaning is clear.

Proposition 31 *Let $\alpha \in \partial_S\mathcal{X}_n$ lie in the boundary component Π, and fix $r \in \mathbb{R}$. Denote the closed horoball and its boundary by*

$$B = \mathrm{B}(\alpha, r), \quad \Sigma = \Sigma(\alpha, r), \quad \overline{\Sigma} = \partial\overline{B}.$$

Then:

- *The horosphere meets the Satake boundary exactly along the closure of the star of Π:*
$$\overline{\Sigma} \cap \partial_S\mathcal{X}_n = \overline{\mathrm{st}(\Pi)}.$$
- *For each Satake point $\beta \in \mathrm{st}(\Pi)$, the hypersurfaces $\overline{\Sigma}$ and $\partial_S\mathcal{X}_n$ are tangent at β.*

Proof **Boundary contact**. Let $\beta \in \partial_S\mathcal{X}_n$. By the asymptotic lemmas of Subsection 2.3:

- If $\beta \in \mathrm{st}(\Pi)$, then $\alpha \in \overline{\mathrm{supp}(\beta)}$, so $\mathfrak{b}_\alpha(\mathbf{Y}) \to -\infty$ as $\mathbf{Y} \to \beta$. Hence $\beta \in \overline{\mathrm{B}(\alpha, r)}$.
- If $\beta \notin \overline{\mathrm{st}(\Pi)}$, then in any neighborhood of β, $\mathfrak{b}_\alpha(\mathbf{Y}) \to +\infty$ along any approach direction, so $\beta \notin \overline{\mathrm{B}(\alpha, r)}$.

This shows

$$\overline{\Sigma} \cap \partial_S\mathcal{X}_n = \overline{B} \cap \partial_S\mathcal{X}_n = \overline{\mathrm{st}(\Pi)}.$$

Tangency. Fix $\beta \in \mathrm{st}(\Pi)$. Conjugate so that

$$\beta = \begin{pmatrix} \beta_0 & 0 \\ 0 & 0 \end{pmatrix}, \quad \beta_0 \in \mathrm{GL}(l, \mathbb{R}).$$

Write a general tangent direction at β in projective coordinates as

$$\mathbf{A} = \begin{pmatrix} \mathbf{A}_1 & \mathbf{A}_2^{\mathsf{T}} \\ \mathbf{A}_2 & \mathbf{A}_3 \end{pmatrix}, \quad \mathbf{A}_3 \in \mathrm{Mat}_{n-l}(\mathbb{R}).$$

With reference point $\mathbf{X} = \mathbf{I}_n$, the horosphere $\Sigma(\alpha, r)$ is cut out (in projective space) by

$$f(\mathbf{Y}) = \big(\mathrm{tr}(\alpha \cdot \mathrm{adj}(\mathbf{Y}))\big)^n - r^n\big(\det \mathbf{Y}\big)^{n-1} = 0,$$

where $\mathrm{adj}(\mathbf{Y})$ denotes the adjugate of $\mathbf{Y}$. Expanding $f(\beta + t\mathbf{A})$ for small t, the dominant term in $r^n(\det(\beta + t\mathbf{A}))^{n-1}$ is

$$r^n(\det(\beta_0)\det(\mathbf{A}_3))^{n-1}t^{(n-1)(n-l)},$$

while $\big(\mathrm{tr}(\alpha \cdot \mathrm{adj}(\beta + t\mathbf{A}))\big)^n$ has strictly higher order in t. Thus $\mathbf{A}$ lies in the tangent cone to $\overline{\Sigma}$ if and only if

$$\det(\mathbf{A}_3) = 0.$$

On the other hand, the Satake boundary $\partial_S \mathcal{X}_n$ in projective coordinates is the hypersurface $\det \mathbf{Y} = 0$. Its linearization at β likewise vanishes on exactly those $\mathbf{A}$ for which $\det(\mathbf{A}_3) = 0$.

Hence at each $\beta \in \mathrm{st}(\Pi)$, the two hypersurfaces $\overline{\Sigma}$ and $\partial_S \mathcal{X}_n$ share the same tangent cone, proving they are tangent. □

We have a more generalized tangency property for non-maximal type cases.

Proposition 32 *Let $\Xi \subset \overline{\mathcal{X}_n}$ be a boundary component of type k, and let $\Pi < \Xi$ be a smaller boundary component containing a Satake point α. For each $r \in \mathbb{R}$, denote the k-th horosphere and its boundary by*

$$B = \mathrm{B}_{\Xi}^{(k)}(\alpha, r), \quad \Sigma = \Sigma_{\Xi}^{(k)}(\alpha, r), \quad \overline{\Sigma} = \partial \overline{B}.$$

Then:

- *The horosphere meets the Satake boundary at:*

$$\overline{\Sigma} \cap \partial_S \mathcal{X}_n = \overline{\mathrm{st}(\Pi) \setminus \Big(\bigsqcup_{\Xi_0 \geq \Xi} \big(\Xi_0 \setminus \mathrm{B}_{\Xi}^{(k)}(\alpha, r)\big)\Big)},$$

  where $\mathrm{B}_{\Xi}^{(k)}(\alpha, r)$ is the corresponding k-th horoball in boundary component Ξ_0.
- *For each Satake point*

$$\beta \in \mathrm{st}(\Pi) \setminus \Big(\bigsqcup_{\Xi_0 \geq \Xi} \big(\Xi_0 \setminus \mathrm{B}_{\Xi}^{(k)}(\alpha, r)\big)\Big),$$

  the hypersurfaces $\overline{\Sigma}$ and $\partial_S \mathcal{X}_n$ are tangent at β.

Proof **Boundary contact**. By the asymptotic lemmas of Subsection 2.3, any Satake point β falls into exactly one of the following cases, determining whether $\beta \in \overline{B}$:

- If $\beta \in \mathrm{st}(\Pi) \setminus \mathrm{st}(\Xi)$, then $\alpha \in \overline{\mathrm{supp}(\beta)}$ and $\Xi \not\leq \mathrm{supp}(\beta)$, so $\mathfrak{b}_{\Xi;\alpha}^{(k)}(\mathbf{Y}) \to -\infty$ as $\mathbf{Y} \to \beta$, and thus $\beta \in B$.
- If $\beta \in \mathrm{st}(\Xi)$, then $\Xi \leq \mathrm{supp}(\beta)$ and $\mathfrak{b}_{\Xi;\alpha}^{(k)}(\mathbf{Y}) \to \mathfrak{b}_{\widetilde{\pi}(\Xi);\widetilde{\pi}(\alpha)}^{(k)}(\widetilde{\pi}(\beta))$ as $\mathbf{Y} \to \beta$, so $\beta \in B$ precisely when that limit is $\leq r$.
- If β is not in the closure of previous cases, $\mathfrak{b}_{\Xi;\alpha}^{(k)}(\mathbf{Y}) \to +\infty$ along any approach, so $\beta \notin \overline{B}$.

To see tangency, fix any such $\beta \in \overline{\Sigma} \cap \partial_S \mathcal{X}_n$. Similar to the previous lemma, a tangent vector $\mathbf{A} \in T_\beta \mathbb{P}(\mathrm{Sym}_n(\mathbb{R}))$ lies in the tangent cone of $\overline{\Sigma}$ if and only if it lies in that of either the equation $\{\mathbf{Y} \mid \det(\pi_\Xi(\mathbf{Y})) = 0\}$ or the boundary defining inequality of $\mathcal{X}_n$. But the latter hypersurface entirely contains $\mathcal{X}_n$, making the two tangent cones coincide. This proves the tangency of $\overline{\Sigma}$ and $\partial_S \mathcal{X}_n$. □

3.2 Satake Face Cycles

As introduced in Lemma 11, the holonomy w of the ideal vertex cycle $\{a_i\}$ preserves the Busemann function based at a. We generalize this to the preservation of Busemann-Selberg functions of all types in the higher-rank setting.

The first proposition concerns maximal-type Busemann-Selberg functions:

Proposition 33 *Let $P = \mathrm{DS}(\mathbf{X}, \Gamma_0)$ be a pre-Dirichlet-Selberg domain with finite faces and finite volume, and is exact with the canonical facet pairing. Let Φ be a Satake face of type k, $\{\Phi_i\}_{i=1}^m$ be a cycle for Φ, and $w \in \mathrm{SL}(n,\mathbb{R})$ be its holonomy.*

Then the restriction of the w-action on $\mathrm{supp}(\Phi)$ has finite order. In addition, there exists a Satake point α_Φ in the relative interior of Φ such that $w.\alpha_\Phi = \alpha_\Phi$, and the maximal type Busemann-Selberg functions based at α_Φ are w-invariant: for every $\mathbf{Y}, \mathbf{Z} \in \mathcal{X}_n$,

$$\mathfrak{b}_{\alpha_\Phi,\mathbf{Z}}(\mathbf{Y}) = \mathfrak{b}_{\alpha_\Phi,\mathbf{Z}}(w.\mathbf{Y}).$$

Proposition 33 rests on the following equivariance lemma.

Lemma 31 *Let $g \in \mathrm{SL}(n,\mathbb{R})$, $\mathbf{X} \in \mathcal{X}_n$, and let $\alpha \in \partial_S \mathrm{Bis}(\mathbf{X}, g^{-1}.\mathbf{X})$. Then:*

1. $\mathrm{tr}(\mathbf{X}^{-1}\alpha) = \mathrm{tr}(\mathbf{X}^{-1}(g.\alpha))$.
2. *For all $\mathbf{Y} \in \mathcal{X}_n$,*
$$\mathfrak{b}_{\alpha,\mathbf{X}}(\mathbf{Y}) = \mathfrak{b}_{g.\alpha,\mathbf{X}}\big(g.\mathbf{Y}\big).$$

Proof Since α lies in the Satake boundary of the bisector $\mathrm{Bis}(\mathbf{X}, g^{-1}.\mathbf{X})$, one has

$$\mathrm{tr}(\mathbf{X}^{-1}\alpha) = \mathrm{tr}((g^{-1}.\mathbf{X})^{-1}\alpha) = \mathrm{tr}(g\mathbf{X}^{-1}g^{\mathsf{T}}\alpha) = \mathrm{tr}(\mathbf{X}^{-1}(g.\alpha)),$$

proving (1).

For (2), note

$$\mathrm{tr}((g.\mathbf{Y})^{-1}(g.\alpha)) = \mathrm{tr}(g^{-1}\mathbf{Y}^{-1}(g^{-1})^{\mathsf{T}}g^{\mathsf{T}}\alpha g) = \mathrm{tr}(g^{-1}\mathbf{Y}^{-1}\alpha g) = \mathrm{tr}(\mathbf{Y}^{-1}\alpha).$$

Hence

$$\mathfrak{b}_{\alpha,\mathbf{X}}(\mathbf{Y}) = \log\frac{\mathrm{tr}(\mathbf{Y}^{-1}(\alpha))}{\mathrm{tr}(\mathbf{X}^{-1}(\alpha))} = \log\frac{\mathrm{tr}((g.\mathbf{Y})^{-1}(g.\alpha))}{\mathrm{tr}(\mathbf{X}^{-1}(g.\alpha))} = \mathfrak{b}_{g.\alpha,\mathbf{X}}(g.\mathbf{Y}).$$

□

Proof (Proposition 33) Recall that the facet pairing $g_{F_i'}$ takes Φ_i to Φ_{i+1}. For this reason, for each $\xi \in \mathrm{span}(\Phi)$, we set $\xi_1 = \xi$ and $\xi_{i+1} = g_{F_i'}.\xi_i$, $i = 1, \ldots, m-1$.

From the canonical facet pairing of pre-Dirichlet-Selberg domains we have $\xi_i \in \mathrm{span}(\Phi_i) \subset \overline{\mathrm{Bis}(\mathbf{X}, g_{F_i}.\mathbf{X})}$. By Lemma 31,

$$\mathrm{tr}(\mathbf{X}^{-1}\xi_i) = \mathrm{tr}(\mathbf{X}^{-1}(g_{F_i'}.\xi_i)) = \mathrm{tr}(\mathbf{X}^{-1}\xi_{i+1}).$$

Iterating gives

$$\mathrm{tr}(\mathbf{X}^{-1}\xi) = \mathrm{tr}(\mathbf{X}^{-1}(w.\xi)). \tag{2}$$

Finite-order on the boundary component. Conjugate so that $\mathrm{supp}(\Phi) = \partial_S(\mathbf{e}_1, \ldots, \mathbf{e}_k)$, and let $\pi : \mathcal{X}_n \sqcup \mathrm{supp}(\Phi) \to \mathcal{X}_k$ be the projection dropping the last $(n-k)$ coordinates (with determinant normalization). Then w preserves both $\mathrm{supp}(\Phi)$ and $\mathrm{span}(\Phi)$, and its restriction $\pi(w) \in \mathrm{GL}(k, \mathbb{R})$ scales to an $\mathcal{X}_k$-isometry.

Define

$$s : \mathrm{span}(\Phi) \to \mathbb{R}, \quad s(\xi) = \frac{\mathrm{tr}(\mathbf{X}^{-1}\xi)}{\det(\iota^{\mathsf{T}}\xi\iota)^{1/k}},$$

where ι is the $n \times k$ matrix selecting the first k coordinates. Then

$$\iota^{\mathsf{T}}(w^{\mathsf{T}}\xi w)\iota = \pi(w)^{\mathsf{T}}(\iota^{\mathsf{T}}\xi\iota)\pi(w).$$

By (2),

$$\begin{aligned} s(w.\xi) &= \frac{\mathrm{tr}(\mathbf{X}^{-1}(w.\xi))}{\det(\pi(w).(\iota^{\mathsf{T}}\xi\iota))^{1/k}} \\ &= \frac{\mathrm{tr}(\mathbf{X}^{-1}\xi)}{|\det(\pi(w))|^{2/k}\det(\iota^{\mathsf{T}}\xi\iota)^{1/k}} = \frac{s(\xi)}{|\det(\pi(w))|^{2/k}}. \end{aligned}$$

On the other hand, by Lemma D1,

$$s(\xi) = \mathrm{tr}\left(\pi(\mathbf{X})^{-1} \cdot \frac{\xi}{\det(\iota^{\mathsf{T}}\xi\iota)^{1/k}}\right) = \mathfrak{s}(\widetilde{\pi}(\mathbf{X}), \pi(\xi))$$

has a unique minimum for $\xi \in \mathrm{span}(\Phi)$, attained when $\pi(\xi)$ is the Selberg nearst-point projection of $\widetilde{\pi}(\mathbf{X})$ to the plane $\pi(\mathrm{span}(\Phi)) \subset \mathcal{X}_k$. Denote this minimum point by α, the uniqueness forces $w.\alpha = \alpha$ and $|\det(\pi(w))| = 1$. Hence $\pi(w)$ lies in the compact subgroup $\mathrm{O}(k)$ and, because it preserves the finitely-sided polytope $\pi(\Phi)$, has finite order.

Existence of a fixed Satake point. If $(\pi(w))^l = \mathbf{I}_k$ for some $l > 0$, then the barycenter of the orbit $\{\xi, \pi(w).\xi, \ldots, \pi(w)^{l-1}.\xi\}$ is a $\pi(w)$-fixed point in the interior of Φ. Lifted back to $\mathcal{X}_n$, this yields the desired α_Φ.

Preservation of the Busemann-Selberg function. Write $\alpha_1 = \alpha_\Phi$ and $\alpha_{i+1} = g_{F_i'}.\alpha_i$. By Lemma 31, for every $\mathbf{Y} \in \mathcal{X}_n$,

$$\mathfrak{b}_{\alpha_i,\mathbf{X}}(\mathbf{Y}) = \mathfrak{b}_{g_{F_i'}.\alpha_i,\mathbf{X}}(g_i.\mathbf{Y}) = \mathfrak{b}_{\alpha_{i+1},\mathbf{X}}(g_i.\mathbf{Y}).$$

Iterating from $i = 1$ to m and using $g_{F_m'}.\alpha_m = \alpha_1$ gives

$$\mathfrak{b}_{\alpha_\Phi,\mathbf{X}}(\mathbf{Y}) = \mathfrak{b}_{\alpha_\Phi,\mathbf{X}}(w.\mathbf{Y}).$$

Replacing $\mathbf{X}$ by any $\mathbf{Z} \in \mathcal{X}_n$ preserves the equality. □

A similar preservation property holds for non-maximal type Busemann-Selberg functions as well.

Proposition 34 *Let $P = \mathrm{DS}(\mathbf{X}, \Gamma_0)$ be a pre-Dirichlet-Selberg domain with finite faces and finite volume, and is exact with the canonical facet pairing. Let Π be a boundary component of type l, $\Psi \subset \Pi$ be a Satake face of the same type.*

Let $\{\Psi\}_{i=1}^m$ be a cycle for Ψ, such that the holonomy w preserves a Satake face $\Phi \subset \partial\Psi$ of type $k < l$ (so $\{\Psi\}_{i=1}^m$ also gives a cycle for Φ). Let $\alpha_\Phi \in \Phi$ be an interior point of Φ fixed by w (cf. Proposition 33). Then for all $\mathbf{Y}, \mathbf{Z} \in \mathcal{X}_n$,

$$\mathfrak{b}^{(l)}_{\Pi;\alpha_\Phi,\mathbf{Z}}(\mathbf{Y}) = \mathfrak{b}^{(l)}_{\Pi;\alpha_\Phi,\mathbf{Z}}(w.\mathbf{Y}).$$

Proof Recall from (1) that

$$\mathfrak{b}^{(l)}_{\Pi;\alpha,\mathbf{Z}}(\mathbf{Y}) = \mathfrak{b}_{\alpha,\mathbf{Z}}(\mathbf{Y}) - \frac{1}{l}\log\det\big(\iota_\Pi^\mathsf{T}\mathbf{Y}^{-1}\iota_\Pi\big),$$

where ι_Π is the $n \times l$ matrix whose columns span Π. By Proposition 33, the usual Busemann-Selberg function $\mathfrak{b}_{\alpha_\Phi,\mathbf{Z}}$ is w-invariant:

$$\mathfrak{b}_{\alpha_\Phi,\mathbf{Z}}(\mathbf{Y}) = \mathfrak{b}_{\alpha_\Phi,\mathbf{Z}}(w.\mathbf{Y}).$$

It remains to check

$$\det(\iota_\Pi^\mathsf{T}\mathbf{Y}^{-1}\iota_\Pi) = \det(\iota_\Pi^\mathsf{T}(w.\mathbf{Y})^{-1}\iota_\Pi).$$

Since w preserves the boundary component Ψ, its action on ι_Π satisfies

$$(w^\mathsf{T})^{-1}\iota_\Pi = \iota_\Pi w', w' \in \mathrm{GL}(l,\mathbb{R}).$$

In fact, matrices ι_Π and $(w^\mathsf{T})^{-1}\iota_\Pi$ represent boundary components Ψ and $w^{-1}.\Psi$, which are assumed to be the same component. Therefore,

$$\begin{aligned}&\det(\iota_\Pi^\mathsf{T}(w.\mathbf{Y})^{-1}\iota_\Pi) = \det(((w^\mathsf{T})^{-1}\iota_\Pi)^\mathsf{T}\mathbf{Y}^{-1}((w^\mathsf{T})^{-1}\iota_\Pi))\\ &= \det((\iota_\Pi w')^\mathsf{T}\mathbf{Y}^{-1}(\iota_\Pi w')) = \det(w')^2\det(\iota_\Pi^\mathsf{T}\mathbf{Y}^{-1}\iota_\Pi).\end{aligned}$$

Finally, Proposition 33 ensures that $\pi_\Pi(w)$ has finite order, so $\det(w')^2 = 1$. Therefore the two determinants agree, and the l-th Busemann-Selberg function is w-invariant. □

3.3 Riemannian Dihedral Angles in Dirichlet-Selberg Domains

For Dirichlet domains in hyperbolic spaces, a critical property is the independence of Riemannian dihedral angles from base point choices. While this fails for Dirichlet-Selberg domains in $\mathcal{X}_n$, understanding the dependence of this angle on the choice of base point is crucial for the proof of the main theorem.

As we defined earlier, a plane $H \subset \mathcal{X}_n$ of codimension k is a non-empty intersection of k linearly-independent hyperplanes. In addition, each of these

hyperplanes is a perpendicular plane for an indefinite matrix $\mathbf{A} \in \mathrm{Sym}_n(\mathbb{R})$. Therefore, the plane can be described as

$$H = \left(\bigcap_{i=1}^{k} \mathbf{A}_i^{\perp} \right) = \mathrm{span}(\mathbf{A}_1, \ldots, \mathbf{A}_k)^{\perp}.$$

In a Dirichlet-Selberg domain, a pair of adjacent faces of codimension k spans two planes H and H' that intersect along $H \cap H'$ of codimension $k+1$. They can be described as

$$H = \mathrm{span}(\mathbf{A}_1, \ldots, \mathbf{A}_{k-1}, \mathbf{B})^{\perp}, \; H' = \mathrm{span}(\mathbf{A}_1, \ldots, \mathbf{A}_{k-1}, \mathbf{B}')^{\perp}, \tag{3}$$

for linearly independent indefinite matrices $\mathbf{A}_1, \ldots, \mathbf{A}_{k-1}, \mathbf{B}, \mathbf{B}' \in \mathrm{Sym}_n(\mathbb{R})$.

Lemma 32 *Let H and H' be planes described as in* (3)*. Then, for any point $\mathbf{X} \in H \cap H'$, the Riemannian dihedral angle $\angle_{\mathbf{X}}(H, H')$ is given by:*

$$\angle_{\mathbf{X}}(H, H') = \arccos \frac{\left(\left(\bigwedge_{i=1}^{k-1} \mathbf{A}_i \right) \wedge \mathbf{B}, \left(\bigwedge_{i=1}^{k-1} \mathbf{A}_i \right) \wedge \mathbf{B}' \right)_{\mathbf{X}^{-1}}}{\sqrt{\left\| \left(\bigwedge_{i=1}^{k-1} \mathbf{A}_i \right) \wedge \mathbf{B} \right\|_{\mathbf{X}^{-1}} \cdot \left\| \left(\bigwedge_{i=1}^{k-1} \mathbf{A}_i \right) \wedge \mathbf{B}' \right\|_{\mathbf{X}^{-1}}}},$$

where $(\cdot,\cdot)_{\mathbf{X}^{-1}}$ denotes the inner product, and $\|\cdot\|_{\mathbf{X}^{-1}}$ the norm, on the exterior algebra $\bigwedge^k(\mathrm{Sym}_n(\mathbb{R}))$ induced by the inner product on $\mathrm{Sym}_n(\mathbb{R})$:

$$\langle \mathbf{A}_1, \mathbf{A}_2 \rangle_{\mathbf{X}^{-1}} = \mathrm{tr}(\mathbf{X}\mathbf{A}_1\mathbf{X}\mathbf{A}_2), \; \forall \mathbf{A}_1, \mathbf{A}_2 \in \mathrm{Sym}_n(\mathbb{R}).$$

Proof In the hypersurface model, the tangent space $T_{\mathbf{X}}H$ is a subspace of $T_{\mathbf{X}}\mathbb{R}^{n(n+1)/2}$:

$$T_{\mathbf{X}}H = \left\{ \mathbf{C} \in T_{\mathbf{X}}\mathbb{R}^{n(n+1)/2} \;\middle|\; \mathrm{tr}(\mathbf{A}_i\mathbf{C}) = 0, \; \mathrm{tr}(\mathbf{B}\mathbf{C}) = 0, \; \mathrm{tr}(\mathbf{X}^{-1}\mathbf{C}) = 0 \right\}.$$

Similarly:

$$T_{\mathbf{X}}H' = \left\{ \mathbf{C} \in T_{\mathbf{X}}\mathbb{R}^{n(n+1)/2} \;\middle|\; \mathrm{tr}(\mathbf{A}_i\mathbf{C}) = 0, \; \mathrm{tr}(\mathbf{B}'\mathbf{C}) = 0, \; \mathrm{tr}(\mathbf{X}^{-1}\mathbf{C}) = 0 \right\}.$$

Recall that the dihedral angles between linear subspaces of $T_{\mathbf{X}}\mathbb{R}^{n(n+1)/2}$ are measured by the inner product given by the Killing form:

$$\langle \mathbf{C}, \mathbf{C}' \rangle_{\mathbf{X}} = \mathrm{tr}(\mathbf{X}^{-1}\mathbf{C}\mathbf{X}^{-1}\mathbf{C}').$$

Thus, the dihedral angle between $T_{\mathbf{X}}H$ and $T_{\mathbf{X}}H'$ is equal to their orthogonal complements with respect to $\langle -, - \rangle_{\mathbf{X}}$. These can be expressed explicitly in terms of bases:

$$(T_{\mathbf{X}}H)^{\perp} = \mathrm{span}(\mathbf{X}, \mathbf{X}\mathbf{A}_1\mathbf{X}, \ldots, \mathbf{X}\mathbf{A}_{k-1}\mathbf{X}, \mathbf{X}\mathbf{B}\mathbf{X}),$$
$$(T_{\mathbf{X}}H')^{\perp} = \mathrm{span}(\mathbf{X}, \mathbf{X}\mathbf{A}_1\mathbf{X}, \ldots, \mathbf{X}\mathbf{A}_{k-1}\mathbf{X}, \mathbf{X}\mathbf{B}'\mathbf{X}).$$

The angle between these complementary spaces is then given by

$$\arccos\left(\det\begin{pmatrix}\langle \mathbf{X}\mathbf{A}_i\mathbf{X}, \mathbf{X}\mathbf{A}_j\mathbf{X}\rangle_{\mathbf{X}} & \langle \mathbf{X}\mathbf{A}_i\mathbf{X}, \mathbf{X}\mathbf{B}\mathbf{X}\rangle_{\mathbf{X}} & \langle \mathbf{X}\mathbf{A}_i\mathbf{X}, \mathbf{X}\rangle_{\mathbf{X}} \\ \langle \mathbf{X}\mathbf{B}'\mathbf{X}, \mathbf{X}\mathbf{A}_j\mathbf{X}\rangle_{\mathbf{X}} & \langle \mathbf{X}\mathbf{B}'\mathbf{X}, \mathbf{X}\mathbf{B}\mathbf{X}\rangle_{\mathbf{X}} & \langle \mathbf{X}\mathbf{B}'\mathbf{X}, \mathbf{X}\rangle_{\mathbf{X}} \\ \langle \mathbf{X}, \mathbf{X}\mathbf{A}_j\mathbf{X}\rangle_{\mathbf{X}} & \langle \mathbf{X}, \mathbf{X}\mathbf{B}\mathbf{X}\rangle_{\mathbf{X}} & \langle \mathbf{X}, \mathbf{X}\rangle_{\mathbf{X}}\end{pmatrix}_{1\le i,j\le k-1}\Bigg/ \left(\det\begin{pmatrix}\langle \mathbf{X}\mathbf{A}_i\mathbf{X}, \mathbf{X}\mathbf{A}_j\mathbf{X}\rangle_{\mathbf{X}} & \langle \mathbf{X}\mathbf{A}_i\mathbf{X}, \mathbf{X}\mathbf{B}\mathbf{X}\rangle_{\mathbf{X}} & \langle \mathbf{X}\mathbf{A}_i\mathbf{X}, \mathbf{X}\rangle_{\mathbf{X}} \\ \langle \mathbf{X}\mathbf{B}\mathbf{X}, \mathbf{X}\mathbf{A}_j\mathbf{X}\rangle_{\mathbf{X}} & \langle \mathbf{X}\mathbf{B}\mathbf{X}, \mathbf{X}\mathbf{B}\mathbf{X}\rangle_{\mathbf{X}} & \langle \mathbf{X}\mathbf{B}\mathbf{X}, \mathbf{X}\rangle_{\mathbf{X}} \\ \langle \mathbf{X}, \mathbf{X}\mathbf{A}_j\mathbf{X}\rangle_{\mathbf{X}} & \langle \mathbf{X}, \mathbf{X}\mathbf{B}\mathbf{X}\rangle_{\mathbf{X}} & \langle \mathbf{X}, \mathbf{X}\rangle_{\mathbf{X}}\end{pmatrix}_{1\le i,j\le k-1}\right.\right.$$
$$\left.\left.\cdot\ \det\begin{pmatrix}\langle \mathbf{X}\mathbf{A}_i\mathbf{X}, \mathbf{X}\mathbf{A}_j\mathbf{X}\rangle_{\mathbf{X}} & \langle \mathbf{X}\mathbf{A}_i\mathbf{X}, \mathbf{X}\mathbf{B}'\mathbf{X}\rangle_{\mathbf{X}} & \langle \mathbf{X}\mathbf{A}_i\mathbf{X}, \mathbf{X}\rangle_{\mathbf{X}} \\ \langle \mathbf{X}\mathbf{B}'\mathbf{X}, \mathbf{X}\mathbf{A}_j\mathbf{X}\rangle_{\mathbf{X}} & \langle \mathbf{X}\mathbf{B}'\mathbf{X}, \mathbf{X}\mathbf{B}'\mathbf{X}\rangle_{\mathbf{X}} & \langle \mathbf{X}\mathbf{B}'\mathbf{X}, \mathbf{X}\rangle_{\mathbf{X}} \\ \langle \mathbf{X}, \mathbf{X}\mathbf{A}_j\mathbf{X}\rangle_{\mathbf{X}} & \langle \mathbf{X}, \mathbf{X}\mathbf{B}'\mathbf{X}\rangle_{\mathbf{X}} & \langle \mathbf{X}, \mathbf{X}\rangle_{\mathbf{X}}\end{pmatrix}_{1\le i,j\le k-1}\right)^{1/2}\right).$$

To simplify this expression, note that

$$\begin{aligned}&\langle \mathbf{X}\mathbf{A}_i\mathbf{X}, \mathbf{X}\mathbf{A}_j\mathbf{X}\rangle_{\mathbf{X}} = \operatorname{tr}(\mathbf{X}^{-1}\mathbf{X}\mathbf{A}_i\mathbf{X}\mathbf{X}^{-1}\mathbf{X}\mathbf{A}_j\mathbf{X})\\ &= \operatorname{tr}(\mathbf{X}\mathbf{A}_i\mathbf{X}\mathbf{A}_j) = \langle \mathbf{X}_i, \mathbf{X}_j\rangle_{\mathbf{X}^{-1}}.\end{aligned}$$

Additionally, since $\mathbf{X} \in H \cap H'$, we have that

$$\langle \mathbf{X}, \mathbf{X}\mathbf{A}_i\mathbf{X}\rangle_{\mathbf{X}} = \operatorname{tr}(\mathbf{A}_i\mathbf{X}) = 0,\ \langle \mathbf{X}, \mathbf{X}\mathbf{B}\mathbf{X}\rangle_{\mathbf{X}} = 0,\ \langle \mathbf{X}, \mathbf{X}\mathbf{B}'\mathbf{X}\rangle_{\mathbf{X}} = 0,$$

and

$$\langle \mathbf{X}, \mathbf{X}\rangle_{\mathbf{X}} = \operatorname{tr}(\mathbf{I}_n) = n.$$

These simplify the formula into the form as presented in Lemma 32. □

An essential corollary of Lemma 32 is the following asymptotic behavior of Riemannian dihedral angles to the Satake boundary:

Proposition 35 *Suppose that H and H' are planes of the same dimension in $\mathcal{X}_n$, and $H \cap H'$ is of codimension 1 in both H and H'. Assume further that Π is a Satake plane of type k in $\overline{\mathcal{X}_n}$, and is transverse to both $\overline{H}$ and $\overline{H'}$. Then for each $\alpha \in \overline{H} \cap \overline{H'} \cap \Pi$ and $\mathbf{Y} \in H \cap H'$, the limit of Riemannian dihedral angle*

$$\lim_{\epsilon \to 0_+} \angle_{\alpha+\epsilon\mathbf{Y}}(H, H') = \angle_{\pi(\alpha)}(\pi(\overline{H} \cap \Pi), \pi(\overline{H'} \cap \Pi)).$$

Here, π is the diffeomorphism from Π to $\mathcal{X}_k$ given in Definition 22.

Proof Without loss of generality, let $\Pi = \partial_S(\mathbf{e}_1, \ldots, \mathbf{e}_k)$, and let

$$H = \operatorname{span}(\mathbf{A}_1, \ldots, \mathbf{A}_{l-1}, \mathbf{B})^{\perp},\ H' = \operatorname{span}(\mathbf{A}_1, \ldots, \mathbf{A}_{l-1}, \mathbf{B}')^{\perp},$$

For $i = 1, \ldots, l-1$, denote the minors of the first k rows and columns of $\mathbf{A}_i$, $\mathbf{B}$ and $\mathbf{B}'$ by $\mathbf{A}_{i,0}$, $\mathbf{B}_0$, and $\mathbf{B}'_0$, respectively. Then,

$$\begin{aligned}\pi(\overline{H} \cap \Pi) &= \operatorname{span}(\mathbf{A}_{1,0}, \ldots, \mathbf{A}_{l-1,0}, \mathbf{B}_0)^{\perp},\\ \pi(\overline{H'} \cap \Pi) &= \operatorname{span}(\mathbf{A}_{1,0}, \ldots, \mathbf{A}_{l-1,0}, \mathbf{B}'_0)^{\perp}.\end{aligned}$$

The transversality of Π to $\overline{H}$ and $\overline{H'}$ ensures that $\mathbf{A}_{0,1}, \ldots, \mathbf{A}_{0,l-1}$, $\mathbf{B}_0$ and $\mathbf{B}_0'$ are linearly independent. By Lemma 32, we have

$$\angle_{\pi(\alpha)}(\pi(\overline{H}\cap\Pi),\pi(\overline{H'}\cap\Pi)) = \arccos\frac{\left(\bigwedge_{i=1}^{l-1}\mathbf{A}_{i,0}\wedge\mathbf{B}_0, \bigwedge_{i=1}^{l-1}\mathbf{A}_{i,0}\wedge\mathbf{B}_0'\right)_{\alpha_0^{-1}}}{\sqrt{\left\|\bigwedge_{i=1}^{l-1}\mathbf{A}_{i,0}\wedge\mathbf{B}_0\right\|_{\alpha_0^{-1}}\cdot\left\|\bigwedge_{i=1}^{l-1}\mathbf{A}_{i,0}\wedge\mathbf{B}_0'\right\|_{\alpha_0^{-1}}}},$$

where $\alpha = \mathrm{diag}(\alpha_0, \mathbf{O})$, i.e., $\alpha_0 = \pi(\alpha)$ is the minor consisting of the first k rows and columns of α. This suggests that

$$\alpha^{1/2}\mathbf{A}_i\alpha^{1/2} = \mathrm{diag}(\alpha_0^{1/2}\mathbf{A}_{i,0}\alpha_0^{1/2}, \mathbf{O}),\ i = 1, \ldots, l-1.$$

Hence, as $\epsilon \to 0$, the inner products for the Riemannian angle have the following limits:

$$\begin{aligned}
&\lim_{\epsilon\to 0}\langle\mathbf{A}_i,\mathbf{A}_j\rangle_{(\alpha+\epsilon\mathbf{Y})^{-1}} = \mathrm{tr}(\alpha\mathbf{A}_i\alpha\mathbf{A}_j)\\
&= \mathrm{tr}((\alpha^{1/2}\mathbf{A}_i\alpha^{1/2})(\alpha^{1/2}\mathbf{A}_j\alpha^{1/2})) = \mathrm{tr}((\alpha_0^{1/2}\mathbf{A}_{i,0}\alpha_0^{1/2})(\alpha_0^{1/2}\mathbf{A}_{j,0}\alpha_0^{1/2}))\\
&= \mathrm{tr}(\alpha_0\mathbf{A}_{i,0}\alpha_0\mathbf{A}_{j,0}) = \langle\mathbf{A}_{i,0},\mathbf{A}_{j,0}\rangle_{\alpha_0^{-1}}.
\end{aligned}$$

By substituting these limits into the expression of $\angle_{\alpha+\epsilon\mathbf{Y}}(H,H')$, we obtain that

$$\lim_{\epsilon\to 0_+}\angle_{\alpha+\epsilon\mathbf{Y}}(H,H') = \angle_{\pi(\alpha)}(\pi(\overline{H}\cap\Pi),\pi(\overline{H'}\cap\Pi)).$$

□

Example 31 *If $H = \mathbf{A}^\perp$ and $H' = \mathbf{B}^\perp$ are hyperplanes, then the Riemannian dihedral angle at any $\mathbf{X} \in H \cap H'$ is given as*

$$\angle_{\mathbf{X}}(H,H') = \arccos\frac{\mathrm{tr}(\mathbf{XAXB})}{\sqrt{\mathrm{tr}((\mathbf{XA})^2)\mathrm{tr}((\mathbf{XB})^2)}}.$$

Let $\mathbf{A} = \mathrm{diag}(\mathbf{A}_0, 0)$ and $\mathbf{B} = \mathrm{diag}(\mathbf{B}_0, 0)$, where

$$\mathbf{A}_0 = \begin{pmatrix} 0 & -1 \\ -1 & 1 \end{pmatrix},\ \mathbf{B} = \begin{pmatrix} 1 & -1 \\ -1 & 0 \end{pmatrix}$$

Then, $\mathbf{A}_0^\perp = \overline{\mathbf{A}^\perp}\cap\partial_S(\mathbf{e}_1,\mathbf{e}_2)$ and $\mathbf{B}_0^\perp = \overline{\mathbf{B}^\perp}\cap\partial_S(\mathbf{e}_1,\mathbf{e}_2)$ are identified with geodesics in $\mathbb{H}^2$, meeting at the point

$$\alpha_0 = \begin{pmatrix} 1 & 1/2 \\ 1/2 & 1 \end{pmatrix},$$

with a Riemannian angle of $2\pi/3$. By Proposition 35, for any line in $H\cap H'$ that tends to $\alpha = \mathrm{diag}(\alpha_0, 0) \in \partial_S\mathcal{X}_3$, the Riemannian dihedral angle between H and H' based at a point on this line will converge to $2\pi/3$, when the base point tends to α.

4 Proof of the Main Theorem

Let $P = \mathrm{DS}(\mathbf{X}, \Gamma) \subset \mathcal{X}_3$ be an exact, finitely-sided Dirichlet-Selberg domain of finite volume satisfying the tiling condition. Recally that P has up to finitely many Satake boundary components of type two, and these components meet only at certain Satake vertices of type one. Let $M = P/\sim$ be the quotient space by gluing along the canonical facet pairing, and $q : P \to M$ be the quotient map. We prove Theorem 11 - the completeness of M - in two steps.

In Subsection 4.1, we construct a subset

$$P^{(1)} \subset P,$$

which is a disjoint union of small neighborhoods around each Satake vertex of type one, such that $P^{(1)}$ meets only the faces incident to those vertices. We then show there exists $r_1 > 0$ so that for every $\mathbf{X} \in P^{(1)}$, the r_1-ball centered at its image $q(\mathbf{X}) \in M$ is complete. Remove these neighborhoods from M, we obtain a manifold (or orbifold) with boundary, denoted by M'. Let $P' \subset P$ be the preimage of M'. By construction, the Satake boundary components of type two in P' are now pairwise disjoint.

In Subsection 4.2, we similarly define

$$P^{(2)} \subset P'$$

as a disjoint union of neighborhoods around each remaining boundary component of type two, meeting only their incident faces. We then prove there exists $r_2 > 0$ so that for all $\mathbf{X} \in P^{(2)} \subset P'$, the r_2-ball around its image in M' is complete. Since the complement $P \setminus (P^{(1)} \cup P^{(2)})$ is bounded, it follows that $M = P/\sim$ is complete.

Throughout the proof we assume, without loss of generality, that the domain P is centered at $\mathbf{X} = \mathbf{I}_3$, the identity matrix.

4.1 Completeness Near Satake Vertices of Type One

We begin by analyzing the cycle structure of Satake vertices of type 1. Since Busemann-Selberg functions depend on chosen reference points, we select them so as to satisfy a natural **vertex-cycle condition**:

Lemma 41 *Let $\alpha \in \partial_S P$ be a Satake vertex of type 1, and let Φ be a Satake face of type 2 containing α. Denote by η and η' the two edges of Φ meet at α, and the boundary component $\Pi = \mathrm{supp}(\Phi)$ of type 2.*

Let $\{\eta_i\}_{i=1}^{m+1}$ be a sequence of Satake edges: $\eta_1 = \eta$, and each η_i is paired with η_{i+1} by the canonical facet pairing of P, and $\eta_{m+1} = \eta'$. Let w be the corresponding word of the facet pairings, similarly to the definition of a holonomy for a Satake cycle, so $w.\eta = \eta'$. Suppose $w.\alpha = \alpha$.

Then there exists a constant $C \geq 0$, depending only on P and α, such that for all $\mathbf{Y} \in \mathcal{X}_3$,

$$|\mathfrak{b}^{(2)}_{\Pi;\alpha,\mathbf{X}}(\mathbf{Y}) - \mathfrak{b}^{(2)}_{\Pi;\alpha,\mathbf{X}}(w.\mathbf{Y})| \leq C.$$

Proof Suppose $\{\eta_i\}_{i=1}^{m+1}$ and $\{\eta'_i\}_{i=1}^{m'+1}$ are sequences such that $\eta_1 = \eta'_1 = \eta$ and $\eta_{m+1} = \eta'_{m'+1} = \eta'$, corresponding to words w and w'. Then the concatenation of the two sequences gives a cycle of η', as well as a cycle of α, with the holonomy $w_0 = w'^{-1}w$.

Applying Proposition 34 to this η'-cycle that preserves α, it follows

$$\mathfrak{b}^{(2)}_{\Pi;\alpha,\mathbf{X}}(w_0.\mathbf{Y}) = \mathfrak{b}^{(2)}_{\Pi;\alpha,\mathbf{X}}(\mathbf{Y}).$$

Substituting $\mathbf{Y}$ with $w'.\mathbf{Y}$ and use $w_0 = w'^{-1}w$:

$$\mathfrak{b}^{(2)}_{\Pi;\alpha,\mathbf{X}}(w.\mathbf{Y}) = \mathfrak{b}^{(2)}_{\Pi;\alpha,\mathbf{X}}(w'.\mathbf{Y}).$$

Since both w and w' preserves α and $\Pi = \mathrm{supp}(\Phi) = \mathrm{supp}(\eta)$, they act on the Busemann-Selberg function $\mathfrak{b}^{(2)}_{\Pi;\alpha,\mathbf{X}}$ by the same constant shift:

$$\mathfrak{b}^{(2)}_{\Pi;\alpha,\mathbf{X}}(w.\mathbf{Y}) = \mathfrak{b}^{(2)}_{\Pi;\alpha,\mathbf{X}}(\mathbf{Y}) + C_{\alpha,\Phi},$$

for certain constant $C_{\alpha,\Phi}$ only depend on Φ, and similarly

$$\mathfrak{b}^{(2)}_{\Pi;\alpha,\mathbf{X}}(w^{-1}.\mathbf{Y}) = \mathfrak{b}^{(2)}_{\Pi;\alpha,\mathbf{X}}(\mathbf{Y}) - C_{\alpha,\Phi}.$$

Since α has a finite orbit under the facet pairing, and each vertex α_i in this orbit is a vertex of finitely many Satake faces Φ_i, we may set

$$C = \max_{\alpha_i,\Phi_i}\{|C_{\alpha_i,\Phi_i}|\},$$

which yields the desired uniform bound. □

Proposition 33, 34 and Lemma 41 implies that for the gluing map $q : P \to M$, we can choose appropriate reference points and have Busemann-Selberg functions $\mathfrak{b}_{\alpha_i,\mathbf{Z}_i}$ and $\mathfrak{b}^{(1)}_{\Pi_i;\alpha_i,\mathbf{Z}'_i}$ for each α_i in the equivalent class $q^{-1}(q(\alpha))$, such that:

- If α_i, α_j are in a certain vertex cycle, and w is the corresponding word of facet pairing transformations so that $\alpha_j = w.\alpha_i$, then

$$\mathfrak{b}_{\alpha_i,\mathbf{Z}_i}(\mathbf{Y}) = \mathfrak{b}_{\alpha_j,\mathbf{Z}_j}(w.\mathbf{Y}), \text{ for any } \mathbf{Y} \in \mathcal{X}_3.$$

- If η_i, η_j are edges in the same edge orbit $q^{-1}q(\eta))$, with $\eta_j = w.\eta_i$ and corresponding boundary components Π_i, Π_j, then

$$|\mathfrak{b}^{(2)}_{\Pi_i;\alpha_i,\mathbf{Z}'_i}(\mathbf{Y}) - \mathfrak{b}^{(2)}_{\Pi_j;\alpha_j,\mathbf{Z}'_j}(w.\mathbf{Y})| \leq C.$$

For convenience, we shall now suppress the reference points and simply denote them by $\mathfrak{b}_{\alpha_i}$ and $\mathfrak{b}^{(1)}_{\Pi_i;\alpha_i}$. We denote by $\mathrm{B}(\alpha_i, r)$ and $\mathrm{B}^{(1)}_{\Pi_i}(\alpha_i, r)$ the associated horoballs. Since P has only finitely many faces, we define a neighborhood of α inside P by

$$\mathrm{B}^{(2)}_P(\alpha, r) = \bigcap_{\Phi \ni \alpha} \mathrm{B}^{(2)}_{\mathrm{supp}(\Phi)}(\alpha, r),$$

where the intersection runs over all type-2 Satake faces $\Phi \ni \alpha$.

As the parameter r approaches to $-\infty$, the lemma below implies that the neighborhood $\mathrm{B}^{(2)}_P(\alpha, r)$ shrinks to the Satake vertex α:

Lemma 42 *For any $r \in \mathbb{R}$, the closure $\overline{\mathrm{B}_P^{(2)}(\alpha, r)} \cap \overline{P}$ contains a neighborhood of α within $\overline{P}$. Moreover, the intersection*

$$\bigcap_{m=1}^{\infty} \left(\overline{\mathrm{B}_P^{(2)}(\alpha, -m)} \cap \overline{P} \right) = \{\alpha\}.$$

Proof For the first assertion, we need to show that $\overline{\mathrm{B}_{\mathit{\Pi}}^{(2)}(\alpha, r)}$ contains a neighborhood of α in $\overline{P}$, where $\mathit{\Pi} = \mathrm{supp}(\mathit{\Phi})$ and $\mathit{\Phi}$ is any type 2 Satake face containing α.

To establish this, let S be a sphere in $\mathbb{RP}^5$ centered at α that intersects every face or Satake face of $\overline{P}$ containing α. Then, the convex hull of $\alpha \sqcup \big(S \cap \overline{P}\big)$ contains a neighborhood of α in $\overline{P}$. We aim to show that this neighborhood is contained in $\overline{\mathrm{B}_{\mathit{\Pi}}^{(2)}(\alpha, r)}$ when the radius of S is sufficiently small. This is justified by showing that the line segment from $[\alpha]$ to $[\alpha + \epsilon\mathbf{X}] \in \mathcal{X}_{n,\mathrm{proj}}$ is entirely contained within $\overline{\mathrm{B}_{\mathit{\Pi}}^{(2)}(\alpha, r)}$, where $\mathbf{X}$ is a point in $S \cap \overline{P}$, and $\epsilon > 0$ depends on $\mathbf{X}$. Such points $\mathbf{X}$ can be categorized into three cases:

1. $\mathbf{X} \in P$,
2. $\mathbf{X} \in \mathit{\Phi}$, or
3. $\mathbf{X}$ lies on a type 2 Satake face distinct from $\mathit{\Phi}$.

Case (1): When $\mathbf{X} \in P$, this containment is straightforward.

Case (2): When $\mathbf{X} \in \mathit{\Phi}$, Lemma 24 implies that for any line $\alpha + \epsilon\mathbf{X} + t\mathbf{Y}$ approaching $\alpha + \epsilon\mathbf{X}$ in $\overline{P}$, where $\mathbf{Y} \in \mathcal{X}_3$, the Busemann-Selberg function $\mathfrak{b}_{\mathit{\Pi};\alpha}^{(2)}(\alpha + \epsilon\mathbf{X} + t\mathbf{Y})$ tends to $\mathfrak{b}_{\widetilde{\pi}(\alpha)}(\widetilde{\pi}(\alpha + \epsilon\mathbf{X}))$, a value less than r for sufficiently small $\epsilon > 0$. Proposition 32 then implies that $\alpha + \epsilon\mathbf{X}$ is on the horosphere $\Sigma_{\mathit{\Pi}}^{(2)}(\alpha, r)$. Thus, the segment from α to $\alpha + \epsilon\mathbf{X}$ remains within $\overline{\mathrm{B}_{\mathit{\Pi}}^{(2)}(\alpha, r)}$.

Case (3): When $\mathbf{X}$ is in a type 2 Satake face distinct from $\mathit{\Phi}$, Lemma 23 ensures that the entire line segment from $\mathbf{X}$ to α lies within $\overline{\mathrm{B}_{\mathit{\Pi}}^{(2)}(\alpha, r)}$.

Since $S \cap \overline{P}$ is compact and $\mathfrak{b}_{\mathit{\Pi};\alpha}^{(2)}$ extends continuously to Satake facets in $\partial_S P$ that contain α, we can select ϵ uniformly over all $\mathbf{X} \in S \cap \overline{P}$. Thus, a neighborhood of α is indeed contained in $\overline{\mathrm{B}_{\mathit{\Pi}}^{(2)}(\alpha, r)}$.

For the second assertion, notice that for any $\mathit{\Phi} \ni \alpha$ and $\mathit{\Pi} = \mathrm{supp}(\mathit{\Phi})$, the intersection

$$\bigcap_{m=1}^{\infty} \left(\overline{\mathrm{B}_{\mathit{\Pi}}^{(2)}(\alpha, -m)} \cap \overline{P} \right)$$

excludes all points in P; by Lemma 24, it also excludes all points in the Satake face $\mathit{\Phi}$, except for α itself. Taking the intersection over all type 2 Satake faces $\mathit{\Phi}$ containing α yields:

$$\bigcap_{m=1}^{\infty} \left(\overline{\mathrm{B}_P^{(2)}(\alpha, -m)} \cap \overline{P} \right) = \bigcap_{\mathit{\Phi}} \bigcap_{m=1}^{\infty} \left(\overline{\mathrm{B}_{\mathit{\Pi}}^{(2)}(\alpha, -m)} \cap \overline{P} \right) = \{\alpha\}.$$

□

Lemma 42 ensures the existence of a constant $r \in \mathbb{R}$ such that the sets $\mathrm{B}_P^{(2)}(\alpha, r)$ for all type 1 Satake vertices $\alpha \in \mathsf{F}_S(P)$ form a disjoint union

$$\bigsqcup_{\alpha} \mathrm{B}_P^{(2)}(\alpha, r),$$

consisting of neighborhoods of those type 1 Satake vertices in $\overline{P}$. The second assertion of the lemma further implies that r can be selected such that each of these component is separated from any face not incident with the corresponding Satake vertex.

We still need a lemma concerning certain relationships between type-two horoballs and classic horoballs based at the same Satake vertex:

Lemma 43 *There exists certain constants $r' \in \mathbb{R}$ and $\epsilon > \epsilon' > 0$, such that:*

1. *For each type-2 Satake face $\Phi \ni \alpha$ with $\Pi = \mathrm{supp}(\Phi)$, and for any face $G \in \mathsf{F}(P)$ either disjoint from Π or incident with α precisely along Π, the set*
$$\mathrm{B}(\alpha, r') \backslash \mathrm{B}_{\Pi}^{(2)}(\alpha, r - C - 2\epsilon)$$
lies at distance at least ϵ from G.
2. *If η and η' are the two edges of Φ meeting at α, and $F, F' \in \mathsf{F}(P)$ are faces incident with η and η' precisely along Π respectively, then their intersections with*
$$\mathrm{B}(\alpha, r') \backslash \mathrm{B}_{\Pi}^{(2)}(\alpha, r - C - 2\epsilon),$$
are separated by distance at least ϵ'.
3. *For any two distinct type-2 Satake faces $\Phi, \Phi' \ni \alpha$,*
$$P \cap \mathrm{B}(\alpha, r') \subset P \cap \left(\mathrm{B}_{\Phi}^{(2)}(\alpha, r - C - 2\epsilon) \cup \mathrm{B}_{\Phi'}^{(2)}(\alpha, r - C - 2\epsilon) \right).$$

Proof **(1).** Consider the nested intersections

$$\overline{P} \cap \left(\bigcap_{m=1}^{\infty} \overline{\mathrm{B}(\alpha, -m) \backslash \mathrm{B}_{\Pi}^{(2)}(\alpha, r - C)} \right).$$

Similar to the proof of Lemma 42, this is the complement of a horoball in Φ based at α, so it is disjoint from any face $G \in \mathsf{F}(P)$ either disjoint from Π or incident with α precisely along Π. By the Lipschitz continuity for Busemann-Selberg functions (Proposition 25), for sufficiently small $r' \in \mathbb{R}$ we obtain a uniform buffer of 3ϵ between

$$\mathrm{B}(\alpha, r') \backslash \mathrm{B}_{\Pi}^{(2)}(\alpha, r - C)$$

for all such faces $G \in \mathsf{F}(P)$. This yields our first assertion.

(2). Similarly, for each of the two edges η, η' through α, the infinite intersections

$$\overline{F} \cap \left(\bigcap_{m=1}^{\infty} \overline{\mathrm{B}(\alpha, -m) \backslash \mathrm{B}_{\Pi}^{(2)}(\alpha, r - C - 2\epsilon)} \right)$$

and

$$\overline{F'} \cap \left(\bigcap_{m=1}^{\infty} \overline{\mathrm{B}(\alpha, -m) \backslash \mathrm{B}_{\Pi}^{(2)}(\alpha, r - C - 2\epsilon)} \right)$$

are the complements of a horoball in the Satake edges η and η'. Hence, by shrinking r' if necessary, one finds ϵ' so that the corresponding truncated regions are at least ϵ' apart, proving our second assertion.

(3). Finally, the infinite intersection

$$\overline{P} \cap \left(\bigcap_{m=1}^{\infty} \overline{\mathrm{B}(\alpha, -m)} \right)$$

is the union of all Satake faces containing α. Since every such face is contained in at least one of the two horoballs $\mathrm{B}_{\Phi}^{(2)}(\alpha, r - C - 2\epsilon)$ or $\mathrm{B}_{\Phi'}^{(2)}(\alpha, r - C - 2\epsilon)$, it follows that for sufficiently $r' \in \mathbb{R}$, $P \cap \mathrm{B}(\alpha, r')$ is contained in the union of these two type-two horoballs. □

With constants C, r, r', and ϵ depending only on the Dirichlet-Selberg domain P defined from Lemmas 41 to 43, we are ready to define the set claimed at the beginning:

$$P^{(1)} = \bigcup_{\alpha} \left(\mathrm{B}_{P}^{(2)}(\alpha, r - C - 2\epsilon) \cap \mathrm{B}(\alpha, r' - \epsilon) \right).$$

As the first half of the proof of the main theorem, we will establish the uniform compactness for balls centered in the quotient $q(P^{(1)})$.

Proof (Proof of Theorem 11, first half) We aim to prove that for every $q(\mathbf{X}) \in q(P^{(1)})$, represented by the point

$$\mathbf{X} \in \bigcup_{\alpha} \left(\mathrm{B}_{P}^{(2)}(\alpha, r - C - 2\epsilon) \cap \mathrm{B}(\alpha, r' - \epsilon) \right),$$

the ball $\mathrm{N}(q(\mathbf{X}), \epsilon'/2)$ is compact. Specifically, we will show that for each such $q(\mathbf{X})$, the preimage of $\mathrm{N}(q(\mathbf{X}), \epsilon'/2)$ is contained in the compact region

$$\bigcup_{\alpha} \left(\mathrm{B}_{P}^{(2)}(\alpha, r) \cap \mathrm{B}(\alpha, r') \backslash \mathrm{B}(\alpha, \mathfrak{b}_{\alpha}(\mathbf{X}) - \epsilon') \right).$$

Assume, by way of contradiction, that there exists a (piecewise smooth) curve γ in $q(P)$ of length $\leq \epsilon'/2$, connecting $q(\mathbf{X})$ and another point $q(\mathbf{Y})$, the latter represented by $\mathbf{Y} \in \mathcal{X}_n$, and

$$\mathbf{Y} \notin \bigsqcup_{\alpha} \left(\mathrm{B}_{P}^{(2)}(\alpha, r) \cap \mathrm{B}(\alpha, r') \right),$$

the disjointness is shown in Lemma 42. Up to a sufficiently small perturbation, we further assume that the preimage of the curve γ does not meet any faces of codimension 2 or more, possibly except for the endpoints $\mathbf{X}$ and $\mathbf{Y}$. Therefore, the preimage is contained in a disjoint union of certain neighborhoods of Satake

vertices $\alpha_1,\ldots,\alpha_N$, consisting of a collection of segments glued together by the quotient map. For any point $q(\mathbf{X}_i) \in q(P)$ where two pieces of the preimage are glued together, its preimage consists of two points $\mathbf{X}_i \sim \mathbf{X}_i'$, paired by a certain facet-pairing transformation g_i, in neighborhoods of certain Satake vertices α_{k_i} and $\alpha_{k_{i-1}}$ of type 1, respectively. We call $\mathbf{X}_i$ and $\mathbf{X}_i'$ a pair of glued points in γ.

Consider the first intersection point of γ with the set

$$\partial \bigcup_{\alpha} \left(\mathrm{B}_P^{(2)}(\alpha, r) \cap \mathrm{B}(\alpha, r') \right),$$

which we denote by $q(\mathbf{Z})$, represented by $\mathbf{Z} \in P$. The preimage of the curve connecting $q(\mathbf{X})$ and $q(\mathbf{Z})$ consists of segments $(\mathbf{X}_0, \mathbf{X}_1')$, $(\mathbf{X}_1, \mathbf{X}_2')$,..., $(\mathbf{X}_{m-1}, \mathbf{X}_m')$, where $\mathbf{X}_i \sim \mathbf{X}_i'$ are pairs of glued points, and $\mathbf{X} = \mathbf{X}_0$, $\mathbf{Z} = \mathbf{X}_m'$ for convenience. We analyze two cases for this intersection point:

- The point $\mathbf{Z}$ lies on $\partial\mathrm{B}(\alpha', r')$ for a certain Satake vertex α' of type 1.
- The point $\mathbf{Z}$ lies on $\partial\mathrm{B}_{\Pi'}^{(2)}(\alpha', r)$ for a certain Satake vertex α' of type 1 and a boundary component $\Pi' = \mathrm{supp}(\Phi')$, where Φ' is a Satake face of type 2 containing α'.

Assume that the first case occurs. Lemma 42 implies that the preimage of the curve restricted to $\mathrm{B}_P^{(2)}(\alpha, r) \cap \mathrm{B}(\alpha, r')$ does not intersect any face not meeting α. Therefore, for each pair of glued points $\mathbf{X}_i \sim \mathbf{X}_i'$ in the curve connecting $q(\mathbf{X})$ and $q(\mathbf{Z})$, Proposition 33 implies the equality

$$\mathfrak{b}_{\alpha_{k_i}}(\mathbf{X}_i) = \mathfrak{b}_{\alpha_{k_{i-1}}}(\mathbf{X}_i').$$

Combining this with the Lipschitz condition for Busemann-Selberg functions (Proposition 25), we deduce that

$$\mathfrak{b}_{\alpha'}(\mathbf{Z}) < \mathfrak{b}_{\alpha}(\mathbf{X}) + \epsilon' < r',$$

given that the segments in the preimage of the curve connecting $q(\mathbf{X})$ and $q(\mathbf{Z})$ have a total length less than ϵ'. However, this contradicts the assumption $\mathbf{Z} \in \partial\mathrm{B}(\alpha', r')$.

Now assume that the second case occurs. Let $(\Pi', \alpha') = (\Pi_{k_{m-1}}, \alpha_{k_{m-1}})$, and inductively define that $(\Pi_{k_{i-1}}, \alpha_{k_{i-1}})$ to be the pair of boundary component with Satake vertex taken to $(\Pi_{k_i}, \alpha_{k_i})$ by g_i. Then $\alpha_{k_0} = \alpha$, and Π_{k_0} is one of the boundary components containing α. Denote it by Π, the assumption implies

$$\mathfrak{b}_{\Pi,\alpha}^{(2)}(\mathbf{X}) \leq r - C - 2\epsilon,\ \mathfrak{b}_{\Pi',\alpha'}^{(2)}(\mathbf{X}') = r.$$

Let Φ_{k_i} be the Satake face contained in Π_{k_i}. Since $\mathbf{X}_i$ and $\mathbf{X}_i'$ lie in the interior of facets of P, $g_i.\Phi_{k_{i-1}}$ and Φ_{k_i} share at least a side. According to the choice of type-two Busemann-Selberg functions, their values $\mathfrak{b}_{\Pi_{k_{i-1}},\alpha_{k_{i-1}}}^{(2)}(\mathbf{X}_i')$ and $\mathfrak{b}_{\Pi_{k_i},\alpha_{k_i}}^{(2)}(\mathbf{X}_i)$ differs by a constant shift $\leq C$. Combining this fact with the

Lipschitz condition for type-two Busemann-Selberg functions (Proposition 25), there is a certain $\mathbf{X}_j$ such that

$$r - C - 2\epsilon \leq \mathfrak{b}^{(2)}_{\Pi_{k_j},\alpha_{k_j}}(\mathbf{X}_j) \leq r - \epsilon.$$

The third assertion in Lemma 43 implies that for each α, the union

$$\bigsqcup_{\Pi \ni \alpha} \mathrm{B}(\alpha, r') \backslash \mathrm{B}^{(2)}_{\Pi}(\alpha, r - C - 2\epsilon)$$

is disjoint. The first assertion in Lemma 43 implies that the preimage of the curve from $q(\mathbf{X}_j)$ to $q(\mathbf{Z})$ restricted to the component for Π of the union above does not meet the faces that are not incident with the two edges η and η' along Π. Moreover, the second assertion in Lemma 43 implies that balls centered at points in the cycle of $\mathbf{X}_j$ with radius $\epsilon'/2$ are disjoint and do not intersect facets that are incident with a different Satake line precisely along the corresponding boundary component. Therefore, along the preimage of the curve from $q(\mathbf{X}_j)$ to $q(\mathbf{Z})$, the corresponding facet-pairing transformations compose into a word w, which maps $\Pi_{k_{j-1}}$ to Π_{k_m}, ensuring that $w.\Phi_{k_{j-1}}$ and Φ_{k_m} share at least a side. Consequently, the values $\mathfrak{b}^{(2)}_{\Pi',\alpha'}(\mathbf{Z})$ is strictly less than r, contradicting the assumption that $\mathbf{Z}$ lies on $\partial \mathrm{B}^{(2)}_{\Pi'}(\alpha', r)$.

This completes the proof of the first half of Theorem 11. □

Remark 41 *We can refine the construction by considering smaller neighborhoods of these Satake vertices, still denoted by $P^{(1)}$, such that any points $\mathbf{X}, \mathbf{X}' \in \partial P$ paired by a side pairing transformation are either both included in or excluded from $P^{(1)}$.*

4.2 Completeness Near Satake Faces of Type Two

We have derived a polytope $P' = P \backslash P^{(1)}$ with unpaired boundary components that does not contain Satake vertices of type 1, and contains only disjoint Satake faces of type 2. In this subsection, we proceed to analyze the cycles of these type 2 Satake faces.

The first lemma in this subsection is parallel to Lemma 42 and proved similarly:

Lemma 44 *Let Φ_i be a Satake face of P, and α_{Φ_i} be an interior point of* $\mathrm{span}(\Phi_i)$. *Then, for any $r \in \mathbb{R}$, the closure $\overline{\mathrm{B}(\alpha_{\Phi_i}, r)} \cap \overline{P'}$ contains a neighborhood of Φ_i in $\overline{P'}$. Furthermore,*

$$\bigcap_{m=1}^{\infty} \left(\overline{\mathrm{B}(\alpha_{\Phi_i}, -m)} \cap \overline{P'} \right) = \Phi_i \cap \overline{P'}.$$

If the Satake face Φ_i is 2-dimensional, the proof requires us to decompose the set $\overline{\mathrm{B}(\alpha_{\Phi_i}, r)} \cap \overline{P'}$ into three mutually exclusive parts:

- Points contained in the δ-neighborhood of a face that is incident with a **vertex** of Φ_i precisely along Π_i,
- Points not of the previous type, while contained in the ϵ-neighborhood of a face incident with an **edge** of Φ_i precisely along Π_i, and
- All other points in $\overline{\mathrm{B}(\alpha_{\Phi_i}, r)} \cap \overline{P'}$.

As shown in the following lemmas, we can choose certain constants $\epsilon, \delta > 0$ such that the second part is a disjoint union corresponding to the edges of Φ_i.

Lemma 45 *Let P_1 and P_2 be hyperplanes in $\mathcal{X}_3$ passing through $\mathbf{I}_3$, and let the Riemannian dihedral angle satisfy*

$$0 < \theta_1 \leq \angle_{\mathbf{I}_3}(P_1, P_2) \leq \theta_2 < \pi.$$

Then for each $\delta > 0$, there exists $\epsilon > 0$ depending on δ, θ_1 and θ_2, such that

$$\mathrm{N}(\mathbf{I}_3, 1) \cap \mathrm{N}(P_1, \epsilon) \cap \mathrm{N}(P_2, \epsilon) \subset \mathrm{N}(\mathbf{I}_3, 1) \cap \mathrm{N}(P_1 \cap P_2, \delta).$$

Here, $\mathrm{N}(P, r)$ denotes the r-neighborhood of P in $\mathcal{X}_3$.

Proof Consider the space of all pairs of hyperplanes in $\mathcal{X}_3$ passing through $\mathbf{I}_3$ with topology induced by their normal vectors. There exists a value ϵ satisfying the inclusion condition, depending on the hyperplane pair (P_1, P_2).

This defines a function on the space of hyperplane pairs, which is continuous and is strictly positive whenever the dihedral angle $\angle_{\mathbf{I}_3}(P_1, P_2)$ is bounded away from 0 and π. Since the space of hyperplane pairs is compact, there exists a constant $\epsilon > 0$ such that the inclusion condition holds for all such pairs (P_1, P_2). □

Lemma 46 *Let η and η' be adjacent edges of the Satake face Φ, such that $\eta \cap \eta' = \alpha$. Let F and F' be faces of P incident with η and η' precisely along* $\mathrm{supp}(\Phi)$*, respectively. Then there is a certain $r \in \mathbb{R}$, such that for every sufficiently small $\delta > 0$, there is a certain $\epsilon > 0$, satisfying*

$$(\mathrm{B}(\alpha_\Phi, r) \cap (F \backslash \mathrm{N}(G, \delta))) \cap (\mathrm{B}(\alpha_\Phi, r) \cap (F' \backslash \mathrm{N}(G, \delta))) = \varnothing,$$

and are separated from each other by distance at least ϵ. Here, $G = F \cap F'$ if it is non-empty. If $F \cap F' = \varnothing$, G is an arbitrary face that is incident with α precisely along $\mathrm{supp}(\Phi)$*.*

Proof **Case (1)**. If $F \cap F' = \varnothing$, we have $\overline{F} \cap \overline{F'} \cap \overline{\mathrm{B}(\alpha_\Phi, r)} = \alpha$. For any G incident with α precisely along $\mathrm{supp}(\Phi)$, Lemma 21 implies that the completion $\overline{\mathrm{N}(G, \delta)}$ contains a neighborhood of α in $\overline{P}$. Therefore,

$$\overline{F \backslash \mathrm{N}(G, \delta)} \text{ and } \overline{F' \backslash \mathrm{N}(G, \delta)}$$

does not meet in $\overline{\mathrm{B}(\alpha_\Phi, r)}$, making them of a positive distance away from each other.

Case (2). Suppose $F \cap F'$ is a face of P incident with α precisely along $\mathit{\Pi} = \mathrm{supp}(\mathit{\Phi})$. Without loss of generality, consider the case when F and F' are facets. According to Proposition 35, the angle $\angle_{\mathbf{X}}(F, F')$ satisfies

$$\angle_{\mathbf{X}}(F, F') \to \angle_{\alpha}(\eta, \eta') := \theta,\ 0 < \theta < \pi,$$

as the base point $\mathbf{X} \in F \cap F'$ is asymptotic to α. By Lemma 44, there exists $r \in \mathbb{R}$ such that

$$\frac{\theta}{2} \leq \angle_{\mathbf{X}}(F, F') \leq \frac{\theta + \pi}{2},$$

for all $\mathbf{X} \in F \cap F' \cap \mathrm{B}(\alpha_{\mathit{\Phi}}, r)$.

Now fix $\mathbf{X} \in F \cap F' \cap \mathrm{B}(\alpha_{\mathit{\Phi}}, r)$. There exists $g \in \mathrm{SL}(3, \mathbb{R})$ such that $g.\mathbf{X} = \mathbf{I}_3$. Moreover,

$$\frac{\theta}{2} \leq \angle_{\mathbf{I}_3}(g.F, g.F') = \angle_{\mathbf{X}}(F, F') \leq \frac{\theta + \pi}{2},$$

where $\mathrm{span}(g.F)$ and $\mathrm{span}(g.F')$ are hyperplanes in $\mathcal{X}_3$ passing through $\mathbf{I}_3$. By Lemma 45, there exists $\epsilon > 0$ such that

$$\mathrm{N}(\mathbf{I}_3, 1) \cap \mathrm{N}(g.F, \epsilon) \cap \mathrm{N}(g.F', \epsilon) \subset \mathrm{N}(\mathbf{I}_3, 1) \cap \mathrm{N}(g.F \cap g.F', \delta).$$

Return to the original faces by g^{-1}-action:

$$\mathrm{N}(\mathbf{X}, 1) \cap \mathrm{N}(F, \epsilon) \cap \mathrm{N}(F', \epsilon) \subset \mathrm{N}(\mathbf{X}, 1) \cap \mathrm{N}(F \cap F', \delta).$$

Since the number $\epsilon > 0$ is independent of $\mathbf{X}$, we apply this for all points $\mathbf{X}$ in $\mathbf{X} \in F \cap F' \cap \mathrm{B}(\alpha_{\mathit{\Phi}}, r)$ and deduce

$$\mathrm{B}(\alpha_{\mathit{\Phi}}, r) \cap \mathrm{N}(F \cap F', 1) \cap \mathrm{N}(F, \epsilon) \cap \mathrm{N}(F', \epsilon) \subset \mathrm{N}(F \cap F', \delta).$$

We claim that for any $\mathbf{Y} \in \mathrm{N}(F, \epsilon)$ and $\mathbf{Y}' \in \mathrm{N}(F', \epsilon)$ not contained in $\mathrm{N}(F \cap F', 1)$, the distance $d(\mathbf{Y}, \mathbf{Y}') \geq 2\epsilon$ as well. Assume this is not true, then for $\mathbf{X} \in F \cap F'$ and lines s, s': $[0, 1] \to \mathcal{X}_3$ from $\mathbf{X}$ respectively to $\mathbf{Y}$ and $\mathbf{Y}'$, the distance from $s(t)$ to s' strictly increases as t increases from 0 to 1. However, when $s(t)$ lies in $\mathrm{N}(F \cap F', 1) \backslash \mathrm{N}(F \cap F', \delta + \epsilon)$, its distance to s' is at least 2ϵ, contradicting the assumption $d(\mathbf{Y}, \mathbf{Y}') < 2\epsilon$.

Thus, we may eliminate $\mathrm{N}(F \cap F', 1)$ from the inclusion above, yielding that

$$\mathrm{B}(\alpha_{\mathit{\Phi}}, r) \cap (F \backslash \mathrm{N}(F \cap F', \delta)) \text{ and } \mathrm{B}(\alpha_{\mathit{\Phi}}, r) \cap (F' \backslash \mathrm{N}(F \cap F', \delta))$$

are separated by distance at least ϵ. □

For each two-dimensional Satake face $\mathit{\Phi}$ and one-dimensional Satake edge η in P', Proposition 33 provides fixed points $\alpha_{\mathit{\Phi}}$ and α_{η} under the corresponding Satake cycles. Moreover, we may choose these points and their Busemann-Selberg functions so that, whenever $\mathit{\Phi}_j = w.\mathit{\Phi}_i$ for some word w in the cycle $q^{-1}(q(\mathit{\Phi}))$, one has

$$\alpha_{\mathit{\Phi}_j} = w.\alpha_{\mathit{\Phi}_i} \text{ and } \mathfrak{b}_{\alpha_{\mathit{\Phi}_i}}(\mathbf{Y}) = \mathfrak{b}_{\alpha_{\mathit{\Phi}_j}}(w.\mathbf{Y}), \quad \forall \mathbf{Y} \in \mathcal{X}_3,$$

and similarly for Satake edge and vertex cycles in P'. We will show that there is $r \in \mathbb{R}$ so that

$$P^{(2)} = \bigcup_{\Pi} \mathrm{B}(\alpha_{\Phi_\Pi}, r)$$

is a disjoint union (indexed by the maximal Satake faces Φ_Π lying in each boundary component Π) and that balls in $q(P^{(2)})$ of a uniform radius are compact.

Proof (Proof of Theorem 11, second half) **Step (1).** By the discussion following Lemma 42 and since there are finitely many Satake vertices in P', we can choose $\delta > 0$ and $r' \in \mathbb{R}$ such that both

$$P^{(2),0} = \bigsqcup_{\alpha} \left(\mathrm{B}(\alpha, r') \cap \bigcup_{F_\alpha} \mathrm{N}(F_\alpha, \delta) \right)$$

and the result by replacing δ with 2δ are disjoint unions (over the Satake vertices α). Here F_α ranges over faces incident with α precisely along the supporting boundary component of Satake faces containing α. Moreover, each component $\mathrm{B}(\alpha, r') \cap \bigcup_F \mathrm{N}(F, \delta)$ does not meet any face that is not incident with the corresponding α. By Proposition 33, $\mathfrak{b}_\alpha$ is invariant under the Satake cycle of α, and one shows exactly as in the classical hyperbolic case that balls of radius δ in in $q(P^{(2),0})$ are compact.

Step (2). If α, α' lie in the interior of the same boundary component Π, then

$$|\mathfrak{b}_{\alpha'}(\mathbf{X}) - \mathfrak{b}_\alpha(\mathbf{X})| < C$$

for some $C > 0$ and all $\mathbf{X}$. Using Lemma 46, we choose ϵ and $r'' \in \mathbb{R}$ so both

$$P^{(2),1} = \bigsqcup_{\eta} \left(\mathrm{B}(\alpha_\eta, r'') \cap \bigcup_{F_\eta} \mathrm{N}(F_\eta, \epsilon) \backslash P^{(2),0} \right),$$

and the result by replacing ϵ with 2ϵ are disjoint unions (over Satake edges η). Moreover, each component does not meet any face that is not incident with the corresponding η. For a similar reason, its image in the quotient has compact ϵ-balls.

Step (3). By the same comparability argument, there is $r''' \in \mathbb{R}$ so that

$$P^{(2),2} = \bigsqcup_{\Phi} \left(\mathrm{B}(\alpha_\Phi, r''') \cap \bigcup_{F_\Phi} \mathrm{N}(F_\Phi, \epsilon) \backslash \left(P^{(2),0} \cup P^{(2),1} \right) \right),$$

is a disjoint union and each component does not meet any face that is not incident with the corresponding Φ. For some $\epsilon' > 0$, balls of radius ϵ' in $q(P^{(2),2})$ are compact.

Finally, the compacrability argument allows us to choose $r \in \mathbb{R}$ small enough ensuring

$$P^{(2)} = \bigcup_{\Pi} \mathrm{B}(\alpha_{\Phi_\Pi}, r) \subset P^{(2),0} \cup P^{(2),1} \cup P^{(2),2},$$

so $P^{(2)}$ is a disjoint union with uniformly compact balls in the quotient, completing the proof. □

Combining the constructions of Subsections 4.1 and 4.2 yields the full proof of Theorem 11.

5 Examples of Dirichlet-Selberg Domains

In this section, we exhibit explicit finite-volume, complete $\mathcal{X}_3$-orbifolds by gluing together certain Dirichlet-Selberg domains along facets. We note that Margulis' Arithmeticity Theorem [19] implies that every lattice in $\mathrm{SL}(3,\mathbb{R})$ is arithmetic. For this reason, the examples in this section are drawn from arithmetic constructions.

Example 51 (The Dirichlet-Selberg Domain for $\Gamma_{C_7}(2)$) *Consider an arithmetic lattice contained in* $\mathrm{SL}(3,\mathbb{Z}) < \mathrm{SL}(3,\mathbb{R})$*, via the surjective homomorphism*

$$\varphi : \mathrm{SL}(3,\mathbb{Z}) \to \mathrm{SL}(3,\mathbb{F}_2),$$

which reduces each entry modulo 2*. The group* $\mathrm{SL}(3,\mathbb{F}_2)$ *has order* 168*, and thus contains* 7*-Sylow subgroups isomorphic to* C_7*; fix one of them to be*

$$C_7 = \left\langle \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{pmatrix} \right\rangle < \mathrm{SL}(3,\mathbb{F}_2).$$

Denote its preimage in $\mathrm{SL}(3,\mathbb{Z})$ *by*

$$\Gamma_{C_7}(2) = \varphi^{-1}(C_7) < \mathrm{SL}(3,\mathbb{Z}),$$

which contains the principal congruence subgroup $\Gamma(2) = \ker(\varphi)$ *with index* 7*.*

Generating set. *Consider the matrices*

$$x = \begin{pmatrix} -1 & 1 & 0 \\ -1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \; y = \begin{pmatrix} 0 & 1 & -1 \\ 1 & 0 & 0 \\ 1 & 0 & -1 \end{pmatrix} \in \Gamma_{C_7}(2) - \Gamma(2).$$

It is straightforward to show that they generate each matrix in the standard generating set of $\Gamma(2)$*, namely:*

(i,j)	$\mathbf{I} + 2\mathbf{e}_i \otimes \mathbf{e}_j$	(i,j)	$\mathbf{I} + 2\mathbf{e}_i \otimes \mathbf{e}_j$	i	$-\mathbf{I} + 2\mathbf{e}_i \otimes \mathbf{e}_i$
$(1,2)$	$xyxy^{-2}$	$(2,1)$	$y^{-1}xyxy^{-1}$	1	y^2xy
$(1,3)$	$xyx^{-1}yxy$	$(3,1)$	$x^{-1}yxyxy$	2	x^2y^{-1}
$(2,3)$	$y^{-2}x^{-3}$	$(3,2)$	$x^{-1}y^{-2}x^{-2}$	3	x^2yxy

This implies that

$$\Gamma(2) < \langle x, y \rangle \leq \Gamma_{C_7}(2).$$

Since no groups properly lie between $\Gamma(2)$ *and* $\Gamma_{C_7}(2)$*, we have* $\langle x, y \rangle = \Gamma_{C_7}(2)$*.*

For a cleaner Dirichlet-Selberg domain, we conjugate x and y by

$$g = \begin{pmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{pmatrix} \in \mathrm{GL}(3,\mathbb{R}),$$

and obtain new generators

$$a = gxg^{-1} = \begin{pmatrix} 1/2 & 1/2 & 0 \\ 1/2 & -1/2 & 1 \\ 1/2 & -1/2 & -1 \end{pmatrix},\ b = gyg^{-1} = \begin{pmatrix} -1/2 & 1 & 1/2 \\ -1/2 & -1 & 1/2 \\ 1/2 & 0 & 1/2 \end{pmatrix}.$$

They generate a lattice,

$$\Gamma = \langle a, b\rangle = g\Gamma_{C_7}(2)g^{-1} < \mathrm{SL}(3,\mathbb{R}),$$

which is isomorphic to the original one.

Dirichlet-Selberg Domain. *Consider the set of words in a and b of length at most 2,*

$$\Gamma_2 = \{a, b, a^{-1}, b^{-1}, a^2, ab, ab^{-1}, \dots\}.$$

Our program [9] computes the pre-Dirichlet-Selberg domain $P = \mathrm{DS}(\mathbf{I}, \Gamma_2)$ and returns the facet data encoded by:

```
generator = np.array([[1/2, 1/2, 0],
        [1/2, -1/2, 1],
        [1/2, -1/2, -1]])
generators = [generator, generator[[1, 2, 0]][:, [1, 2, 0]]]
center = np.eye(3)
wbs, faces = compute_selberg_domain(generators, center, 2)
```

The `faces` *output suggests that P is a 5-simplex, while from the facet equations in the* `wbs` *output, we recover the vertices on the Satake boundary:*

$$\alpha_{1,2} = \begin{pmatrix} 1 & \pm 1 & 0 \\ \pm 1 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix},\ \alpha_{3,4} = \begin{pmatrix} 1 & 0 & \pm 1 \\ 0 & 0 & 0 \\ \pm 1 & 0 & 1 \end{pmatrix},\ \alpha_{5,6} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & \pm 1 \\ 0 & \pm 1 & 1 \end{pmatrix}.$$

Under the identification of the type-one component of $\partial_S \mathcal{X}_3$ with $\mathbb{RP}^2$, these correspond to $\alpha_{1,2} = [1 : \pm 1 : 0]$, $\alpha_{3,4} = [1 : 0 : \pm 1]$, and $\alpha_{5,6} = [0 : 1 : \pm 1]$.

For the facets F_i spanned by the five vertices missing α_i, $i = 1, \dots, 6$, the program also suggests

$$F_1 \subset \mathrm{Bis}(\mathbf{I}, a.\mathbf{I}),\ F_3 \subset \mathrm{Bis}(\mathbf{I}, b.\mathbf{I}),\ F_5 \subset \mathrm{Bis}(\mathbf{I}, c.\mathbf{I}),$$

and

$$F_6 \subset \mathrm{Bis}(\mathbf{I}, a^{-1}.\mathbf{I}),\ F_2 \subset \mathrm{Bis}(\mathbf{I}, b^{-1}.\mathbf{I}),\ F_4 \subset \mathrm{Bis}(\mathbf{I}, c^{-1}.\mathbf{I}),$$

where $c = a^{-1}b^{-1}$.

Ridge Cycles and Properness. *To verify the exactness and ridge cycles, we continue to run the code:*

```
pairing_dict = polytope_exactness(wbs, faces)
if pairing_dict is None:
    print("Domain is not exact.")
else:
    cycles = compute_ridge_cycles(wbs, faces)
    for i, cycle in enumerate(cycles):
        k = compute_angle_sum(wbs, faces, cycle)
        print(f"Cycle {i}: ridge indices={cycle.ridge},\
        angle-sum divisor={k}")
```

The output suggests that P is exact with respect to the canonical facet pairing, such as $a.F_6 = F_1$. The fifteen ridges $r_{ij} = F_i \cap F_j$, $1 \le i < j \le 6$, break into five cycles under the side-pairing transformations:

Ridges $\{r_i\}_{i=1}^m$	*Transformations* $\{g_i\}_{i=1}^m$	*Angle sum*	*Relator*
$\{r_{56}, r_{12}, r_{34}\}$	$\{a, b, c\}$	2π	abc
$\{r_{14}, r_{36}, r_{25}\}$	$\{a^{-1}, b^{-1}, c^{-1}\}$	π	$(cba)^2$
$\{r_{26}, r_{16}, r_{13}\}$	$\{a, a, b^{-1}\}$	π	$(a^2b^{-1})^2$
$\{r_{24}, r_{23}, r_{35}\}$	$\{b, b, c^{-1}\}$	π	$(b^2c^{-1})^2$
$\{r_{46}, r_{45}, r_{15}\}$	$\{c, c, a^{-1}\}$	π	$(c^2a^{-1})^2$

Since each angle sum is a submultiple of 2π, the facet pairing for P is proper.

Gluing the facets of P via the identifications in Γ_2 produces a complete, finite-volume $\mathcal{X}_3$-orbifold $M = P/\sim$. By Theorem 11, the orbifold is complete. Poincaré's Fundamental Polyhedron Theorem implies that P is the Dirichlet-Selberg domain $\mathrm{DS}(\mathbf{I}, \Gamma)$ for the group $\Gamma = \langle a, b\rangle \cong \Gamma_{C_7}(2)$. Since conjugation does not affect the abstract group structure, these relators also yield a presentation for $\Gamma_{C_7}(2)$:

$$\Gamma_{C_7}(2) = \langle a, b \mid (aba^{-1}b^{-1})^2, (ababa)^2, (a^2b^{-1})^2, (ab^3)^2\rangle.$$

Example 52 (The Dirichlet-Selberg Domain for $\Gamma'(2)$) *As another example, consider the congruence subgroup*

$$\Gamma = \Gamma'(2) = \{g = (g_{ij})_{i,j=1}^3 \in \mathrm{SL}(3,\mathbb{Z}) \mid g_{ii} \in 1 + 4\mathbb{Z},\ g_{ij} \in 2\mathbb{Z},\ \forall i \ne j\}.$$

This group is generated by the matrices [20, 1]

$$a_{ij} = \mathbf{I} + 2\mathbf{e}_i \otimes \mathbf{e}_j, \quad 1 \le i \ne j \le 3.$$

Dirichlet-Selberg Domain. *Similarly to the previous example, consider the subset Γ_2 of words of length at most 2. To compute the structure of the pre-Dirichlet-Selberg domain $P = \mathrm{DS}(\mathrm{I}, \Gamma_2)$, run the code in the program:*

```
generators = []
for i, j in itertools.permutations(range(3), 2):
    generator = np.eye(3)
    generator[i, j] = 2
    generators.append(generator)
```

```
center = np.eye(3)
wbs, faces = compute_selberg_domain(generators, center, 2)
```

The `faces` *data suggests that the Dirichlet-Selberg domain P has* 24 *facets, contained in the bisectors*

$$\mathrm{Bis}(\mathbf{I}, a_{ij}^{\pm}.\mathbf{I}) \text{ and } \mathrm{Bis}(\mathbf{I}, a_{ji}^{\pm}a_{ki}^{\pm}.\mathbf{I}), \quad i \neq j,\ j \neq k,\ i \neq k.$$

From the `wbs` *data we recover the Satake vertices. All of these are type one, and in the shorthand notation of* $\mathbb{RP}^2$,

$$[1:0:0], [0:1:0], [0:0:1], [1:\pm1:0], [1:0:\pm1], [0:1,\pm1], [1:\pm1:\pm1].$$

Ridge Cycles and Properness. *By running the code same as in the previous example, we see that P is exact with the canonical facet pairing, and its* 84 *ridges are organized into* 25 *ridge cycles of four types. We summarize the ridge cycles as below.*

Quantity	*Transformations*	*Angle sum*	*Relator*
12	$\{a_{ik}, a_{jk}, a_{ik}^{-1}a_{jk}^{-1}\}$	2π	$[a_{ik}, a_{jk}]$
3	$\{a_{ij}, a_{ik}, a_{ij}^{-1}, a_{ik}^{-1}\}$	2π	$[a_{ij}, a_{ik}]$
6	$\{a_{ij}, a_{ik}a_{jk}^{-1}, a_{ij}^{-1}, a_{ik}a_{jk}\}$	2π	$[a_{ij}, a_{jk}]a_{ik}^{-2}$
4	$\{a_{13}^{-1}a_{23}, a_{21}^{-1}a_{31}, a_{32}^{-1}a_{12}\}$[1]	π	$(a_{12}a_{13}^{-1}a_{23}a_{21}^{-1}a_{31}a_{32}^{-1})^2$

Similarly to the previous example, the facet pairing for P is proper, and P is the Dirichlet-Selberg domain for the group $\Gamma = \Gamma'(2)$. We obtain the presentation:

$$\Gamma'(2) = \langle a_{ij} \,|\, [a_{ij}, a_{ik}],\ [a_{ik}, a_{jk}],\ [a_{ij}, a_{jk}]a_{ik}^{-2},\ (a_{12}a_{13}^{-1}a_{23}a_{21}^{-1}a_{31}a_{32}^{-1})^2 \rangle.$$

A An Analyze on Finite-volume Polytopes

We aim to prove Proposition 12, claiming that an $\mathcal{X}_n$-polytope has finite volume if and only if its corresponding projective polytope is contained in the region $\overline{\mathcal{X}_{n,\mathrm{proj}}}$. Before the proof, let us give the explicit volume form on $\mathcal{X}_n$.

Lemma A1 *The volume form on the projective model $\mathcal{X}_{n,proj}$ is*

$$d\mu(x_{ij}) = 2^{n^2/2} n^{n(n+1)/4-1} (\det(x_{ij}))^{-(n+1)/2} \iota_E \bigwedge_{i\leq j} \mathrm{d}x_{ij}, \quad E = \sum_{i\leq j} x_{ij} \frac{\partial}{\partial x_{ij}}.$$

Here, the coordinates x_{ij} are the entries of the matrix $\mathbf{X}$ representing a point $[\mathbf{X}] \in \mathcal{X}_{n,proj}$.

Proof We first compute the volume form on the positive cone

$$\mathcal{P}_n = \mathrm{GL}^+(n,\mathbb{R})/\mathrm{SO}(n) = \{\mathbf{X} \in \mathrm{Sym}_n(\mathbb{R}) \mid \mathbf{X} > 0\},$$

[1] This is one representative example among the four ridge cycles. Using the commuting and Heisenberg-type relators, the relators for all four are shown to be equivalent.

with respect to the metric tensor adopted in [5]:

$$g_{\mathbf{X}}(\mathbf{A},\mathbf{B}) = \operatorname{tr}(\mathbf{X}^{-1}\mathbf{A}\mathbf{X}^{-1}\mathbf{B}).$$

At the identity matrix $\mathbf{I}$, the basis matrices $\mathbf{A}_{ii} = \mathbf{e}_i \otimes \mathbf{e}_i$, $i = 1, \dots, n$, and $\mathbf{A}_{ij} = \mathbf{e}_i \otimes \mathbf{e}_j + \mathbf{e}_j \otimes \mathbf{e}_i$, $1 \le i < j \le n$, are pairwise orthogonal with respect to $g_{\mathbf{I}}$. Therefore, the volume form at $\mathbf{I} \in \mathcal{P}_n$ is

$$\mathrm{d}\mu_{\mathcal{P}}(\mathbf{I}) = \sqrt{\det(g)} \bigwedge_{i\le j} \mathrm{d}x_{ij} = \sqrt{\prod_{i\le j} g_{\mathbf{I}}(\mathbf{A}_{ij},\mathbf{A}_{ij})} \bigwedge_{i\le j} \mathrm{d}x_{ij} = 2^{n(n-1)/4} \bigwedge_{i\le j} \mathrm{d}x_{ij}.$$

A general $\mathbf{X} \in \mathcal{P}_n$ is taken to the identity by the $\mathrm{GL}(n,\mathbb{R})$-action

$$\phi(\mathbf{Y}) = \mathbf{X}^{-1/2}\mathbf{Y}\mathbf{X}^{-1/2}.$$

Since this action preserves the metric, the volume form at $\mathbf{X}$ is the pullback of the volume form at $\mathbf{I}$:

$$\mathrm{d}\mu_{\mathcal{P}}(\mathbf{X}) = \phi^* \mathrm{d}\mu_{\mathcal{P}}(\mathbf{I}) = J_\phi \, 2^{n(n-1)/4} \bigwedge_{i\le j} \mathrm{d}x_{ij}.$$

This $\mathrm{GL}(n,\mathbb{R})$-action is the symmetric square of the $\mathrm{GL}(n,\mathbb{R})$-action on $\mathbb{R}^n$; see [13], Chapter 11. The Jacobian of this map is

$$J_\phi = (\det \mathbf{X}^{-1/2})^{n+1} = \det(\mathbf{X})^{-(n+1)/2}.$$

Hence the symmetric cone $\mathcal{P}_n$ has volume form

$$\mathrm{d}\mu_{\mathcal{P}}(\mathbf{X}) = 2^{n(n-1)/4} (\det(x_{ij}))^{-(n+1)/2} \bigwedge_{i\le j} \mathrm{d}x_{ij}.$$

Now set

$$\tau = \frac{1}{\sqrt{n}} \log\det(\mathbf{X}),$$

so that $\mathcal{P}_n \cong \mathbb{R} \times \mathcal{X}_n$ via [5], Proposition 10.53:

$$(\tau, \mathbf{Y}) \longmapsto e^{\tau/\sqrt{n}}\mathbf{Y}, \quad \det(\mathbf{Y}) = 1.$$

With this normalization, the metric decomposition becomes

$$ds^2_{\mathcal{P}} = d\tau^2 + ds^2_{\mathcal{X}},$$

and hence the volume form on $\mathcal{P}_n$ can be written as

$$d\mu_{\mathcal{P}} = d\tau \wedge d\mu_{\mathcal{X}}.$$

On the other hand, the radial vector field for τ is

$$\partial_\tau = \frac{1}{\sqrt{n}} \sum_{i\le j} x_{ij} \frac{\partial}{\partial x_{ij}} = \frac{1}{\sqrt{n}} E.$$

Therefore, we obtain the volume form on $\mathcal{X}_n$ by contraction with ∂_τ:

$$d\mu_{\mathcal{X}} = \iota_{\partial_\tau}(d\mu_{\mathcal{P}}) = \frac{1}{\sqrt{n}} \iota_E(d\mu_{\mathcal{P}}) = \frac{2^{n(n-1)/4}}{\sqrt{n}} (\det(x_{ij}))^{-(n+1)/2} \iota_E \bigwedge_{i\le j} \mathrm{d}x_{ij}.$$

This volume form is homogeneous of degree 0, so passing from $\mathcal{X}_{n,\mathrm{hyp}}$ to the projective model $\mathcal{X}_{n,\mathrm{proj}}$ does not change its formula.

Finally, scaling the metric tensor by a factor of $2n$ throughout this paper induces an additional factor of $(2n)^{(n^2+n-2)/4}$ in the volume form, leading to the formula stated in the lemma. □

As the constant coefficient $2^{n^2/2}n^{n(n+1)/4-1}$ does not affect our conclusion in Proposition 12, we omit it for simplicity in the proof below.

Proof (Proposition 12) **Necessity**. If $\widehat{P} \not\subset \overline{\mathcal{X}_n}$, then there is a boundary point $\mathbf{X}_0 \in \partial_S P$ with $\det \mathbf{X}_0 = 0$ and a small projective-neighborhood $U \ni \mathbf{X}_0,\ U \subset \widehat{P}$. On $U \cap \partial_S \mathcal{X}_n$ the denominator $\det \mathbf{X}$ vanishes to first order, so $\int_{U\cap\overline{\mathcal{X}_n}} \mathrm{d}\mu = \infty$. Hence P cannot have finite volume.

Sufficiency. Conversely, assume $\widehat{P} \subset \overline{\mathcal{X}_n}$. We show each boundary neighborhood contributes a finite amount to the volume integral; compact interior patches are manifestly finite.

Fix a boundary stratum of type $n-k$. After conjugating, we may take

$$\mathbf{X}_0 = \mathrm{diag}(\mathbf{I}_k, \mathbf{O}), \quad k < n.$$

Introduce homogeneous coordinates near $\mathbf{X}_0$:

$$\mathbf{X} = \mathbf{X}_0 + \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^{\mathsf{T}} & \mathbf{C} \end{pmatrix},$$
$$\mathbf{A} \in \mathrm{Sym}_k(\mathbb{R}),\ \mathrm{tr}(\mathbf{A}) = 0,\ \mathbf{B} \in M_{k,n-k}(\mathbb{R}),\ \mathbf{C} \in \mathrm{Sym}_{n-k}(\mathbb{R}).$$

Positivity of $\mathbf{X}$ forces $\mathbf{C}$ to lie in a convex polytope of the projective cone. Writing $\mathbf{C} = t\,\mathbf{Y}$ with $t \geq 0$ and $\mathbf{Y} \in \mathcal{X}_{n-k}$, then $\mathbf{Y}$ lies in a Satake face P_0 in the boundary component $\mathcal{X}_{n-k}$ consisting of positive-definite lower-right $(n-k)\times(n-k)$ blocks. For fixed $\mathbf{Y} \in P_0$, positive-definiteness along with the polyhedral property together show that entries in $\mathbf{B}$ are $O(t)$, therefore $\det(\mathbf{X}) = t^{n-k}\det(\mathbf{Y}) + O(t^{n-k+1})$. By the compactness of $\overline{P_0}$ one has

$$\det(\mathbf{X}) \geq \frac{1}{2} t^{n-k} \det(\mathbf{Y}),$$

in a sufficiently small neighborhood.

The volume form factorizes (up to a bounded Jacobian) as

$$\iota_E \bigwedge \mathrm{d}x_{ij} \ \sim\ \left(\iota \bigwedge \mathrm{d}a_{ij}\right)\left(\bigwedge \mathrm{d}b_{ij}\right)\left(\bigwedge \mathrm{d}c_{ij}\right),$$

where a_{ij}, b_{ij}, and c_{ij} denotes the entries of $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{C}$, respectively. The operator ι is the interior product corresponding to the restriction $\mathrm{tr}(\mathbf{A}) = 0$.

For fixed $\mathbf{C}$, the positive-definiteness of $\mathbf{X}$ narrows the domain of variables b_{ij} to

$$\int \bigwedge \mathrm{d}b_{ij} \leq 2^{k(n-k)}(\det(\mathbf{C}))^{k/2} = 2^{k(n-k)}(\det(\mathbf{Y}))^{k/2} t^{k(n-k)/2}.$$

For fixed t, $\mathbf{B} = O(t)$, $\int \bigwedge \mathrm{d}b_{ij} = O(t^{k(n-k)})$, thus one estimates

$$\int \bigwedge \mathrm{d}b_{ij} \leq K(\det(\mathbf{Y}))^{k/2} t^{k(n-k)},\ K < \infty.$$

Meanwhile, $\mathbf{C} = t\mathbf{Y}$ contributes

$$\bigwedge \mathrm{d}c_{ij} = t^{(n-k-1)(n-k+2)/2}\mathrm{d}t \wedge \left(\iota_E \bigwedge_{j\geq i\geq k+1} \mathrm{d}y_{ij}\right),\ E = \sum_{i\leq j} y_{ij}\frac{\partial}{\partial y_{ij}}.$$

Hence the singularity near $\mathbf{X}$ is integrable:

$$\begin{aligned}
\int_{U_\epsilon} \mathrm{d}\mu &= \int_{U_\epsilon} \frac{\iota_E \bigwedge \mathrm{d}x_{ij}}{\det(\mathbf{X})^{\frac{n+1}{2}}} \le \int_{U_\epsilon} \frac{2^{\frac{n+1}{2}} \iota_E \bigwedge \mathrm{d}x_{ij}}{\det(t\mathbf{Y})^{\frac{n+1}{2}}} \\
&= \int_{|x_{ij}|<\epsilon} \iota \bigwedge \mathrm{d}a_{ij} \int \bigwedge \mathrm{d}b_{ij} \int_0^\epsilon \mathrm{d}t \int_{P_0} \frac{2^{\frac{n+1}{2}} t^{\frac{(n-k-1)(n-k+2)}{2}} \iota_E \bigwedge_{i\le j} \mathrm{d}y_{ij}}{\det(t\mathbf{Y})^{\frac{n+1}{2}}} \\
&\le (2\epsilon)^{\frac{(k-1)(k+2)}{2}} \int_0^\epsilon \mathrm{d}t \int_{P_0} K(\det(\mathbf{Y}))^{\frac{k}{2}} t^{k(n-k)} \frac{2^{\frac{n+1}{2}} t^{\frac{(n-k-1)(n-k+2)}{2}} \iota_E \bigwedge_{i\le j} \mathrm{d}y_{ij}}{\det(t\mathbf{Y})^{\frac{n+1}{2}}} \\
&= K' \epsilon^{\frac{(k-1)(k+2)}{2}} \int_0^\epsilon \mathrm{d}t \int_{P_0} \frac{t^{\frac{k(n-k)}{2}-1} \iota_E \bigwedge_{i\le j} \mathrm{d}y_{ij}}{\det(\mathbf{Y})^{\frac{n-k+1}{2}}} \\
&= \frac{K' \epsilon^{k(n+1)-1}}{k(n-k)} \mathrm{Vol}(P_0) < \infty,
\end{aligned}$$

using the induction assumption for $\mathcal{X}_{n-k}$. Covering $\widehat{P}$ by finitely many such boundary charts plus an interior compact set shows $\int_P \mathrm{d}\mu < \infty$. □

B Maximal Satake faces of Finite-volume Polytopes

We will prove Proposition 13, stating that the maximal Satake faces of a finite-volume polytope are finite-volume polytopes in the corresponding boundary component.

Proof (Proposition 13) Let $\overline{P} = \mathrm{conv}\{\alpha_1, \dots, \alpha_m\}$, where each vertex $\alpha_i \in \overline{\mathcal{X}_n}$. Let $V \subset \mathbb{R}^n$ be a subspace such that $\Phi_V = \partial_S P \cap \partial_S(V)$ is non-empty, and let

$$\mathsf{I} = \{i \,|\, \alpha_i \in \overline{\partial_S(V)}\}.$$

We claim the following:

- The convex hull $\mathrm{conv}(\{\alpha_i\}_{i\in\mathsf{I}})$ is a face of $\overline{P}$.
- This convex hull coincides with $\overline{\Phi_V}$.

To prove the first claim, recall that $\alpha \in \overline{\partial_S(V)}$ if and only if the associated bilinear form vanishes on $V^\perp$. For each $j \notin \mathsf{I}$, the kernel of α_j in $V^\perp$ is a proper Zariski-closed subset; we may therefore choose $\mathbf{u} \in V^\perp$ and $\epsilon > 0$ such that

$$\mathbf{u}^\mathsf{T} \alpha_i \, \mathbf{u} = 0 \quad (i \in \mathsf{I}), \quad \text{and} \quad \mathbf{u}^\mathsf{T} \alpha_j \, \mathbf{u} \ge \epsilon \quad (j \notin \mathsf{I}).$$

The hyperplane $\{\alpha \mid \mathbf{u}^\mathsf{T} \alpha \, \mathbf{u} = 0\}$ separates $\mathrm{conv}(\{\alpha_i\}_{i\in\mathsf{I}})$ from $\mathrm{conv}(\{\alpha_j\}_{j\notin\mathsf{I}})$, proving that the former is indeed a face.

For the second claim, observe that $\overline{\Phi_V} = \overline{\partial_S(V)} \cap \overline{P}$ is convex and contains all α_i for $i \in \mathsf{I}$, so $\mathrm{conv}(\{\alpha_i\}_{i\in\mathsf{I}}) \subseteq \overline{\Phi_V}$. Conversely, any point of $\overline{\Phi_V}$ is a convex combination of the vertices $\alpha_1, \dots, \alpha_m$. The separation argument above applies to $\overline{\Phi_V}$ as well, implying that vertices with $j \notin \mathsf{I}$ cannot appear in such combinations. Thus, $\overline{\Phi_V} = \mathrm{conv}(\{\alpha_i\}_{i\in\mathsf{I}})$, as claimed. □

C An Inequality for Interlaced Sequence Deviations

In this appendix, we have an inequality used in the proof of Lemma 21 for the Lipschitz continuity of Busemann-Selberg functions.

Lemma C1 *Let n and k be positive integers with $k < n$. Suppose*

$$a_1 \geq a_2 \geq \cdots \geq a_n \quad \textit{and} \quad b_1 \geq b_2 \geq \cdots \geq b_k$$

are real numbers satisfying the interlacing condition,

$$a_i \geq b_i \geq a_{i+n-k},\ i = 1, \ldots, k.$$

Define the averages $\bar{a} = \frac{1}{n}\sum a_i$ and $\bar{b} = \frac{1}{k}\sum b_i$. Then,

$$\sum_{i=1}^{n}(a_i - \bar{a})^2 \geq \sum_{i=1}^{k}(b_i - \bar{b})^2.$$

Proof We proceed by induction on $n - k$. The base case $n - k = 1$ asserts that

$$a_1 \geq b_1 \geq a_2 \geq \cdots \geq b_{n-1} \geq a_n.$$

Fix such an interlacing (a_i) and (b_i). Since the squared deviation

$$\mathbb{R}^{n-1} \to \mathbb{R},\ (b_1, \ldots, b_{n-1}) \mapsto \sum_{i=1}^{n-1}(b_i - \bar{b})^2$$

is convex, its restriction to $[a_2, a_1] \times \cdots \times [a_{n-1}, a_n]$ attains the maximum at a certain corner.

Furthermore, if for any $i \leq j$, $a_i = a_{i+1} = \cdots = a_j$, and $a_{i-1} > a_i$, $a_j > a_{j+1}$ are strict, then $b_{i-1} = b_j = a_i$ is not a maximum: if this holds, then for sufficiently small $\epsilon > 0$, replacing them with $b_{i-1} = a_i + \epsilon$ and $b_j = a_i - \epsilon$ does not change the mean value $\bar{b}$ and increases the deviation $\sum(b_i - b)^2$. Hence up to reordering, the maximum point $\{b_1, \ldots, b_{n-1}\}$ must equal $\{a_1, \ldots, a_n\} \setminus \{a_j\}$ for some $1 \leq j \leq n$.

A direct calculation then shows

$$\sum_{i=1}^{n}(a_i - \bar{a})^2 - \sum_{i \neq j}\left(a_i - \frac{\sum_{i \neq j} a_i}{n-1}\right)^2 = \frac{n-2}{n-1}(a_j - \bar{a})^2 \geq 0,$$

which establishes the case $k = n - 1$.

For general $n - k$, one has a refined sequence $\{b'_1, \ldots, b'_{k+1}\}$ satisfying $b_{i-1} \geq b'_i \geq b_i$ and $a_i \geq b'_i \geq a_{i+n-k-1}$. Applying the induction assumption to $\{a_i\}$ and $\{b'_i\}$ yields the desired inequality. □

D Selberg Nearest-Point Projections to $\mathcal{X}_n$-planes

The result in this appendix is required by Proposition 33 for to the preservation property of Busemann-Selberg functions by the holonomies of Satake face cycles.

Lemma D1 *Let H be a plane in $\mathcal{X}_n$, and let $\mathbf{X} \in \mathcal{X}_n$ be a point. Then the Selberg premetric from $\mathbf{X}$ to H,*

$$\mathbf{Y} \mapsto \mathfrak{s}(\mathbf{X}, \mathbf{Y}) = \operatorname{tr}(\mathbf{X}^{-1}\mathbf{Y}), \quad \mathbf{Y} \in H,$$

attains a unique minimum.

Proof Since $\mathbf{X}$ is represented by a positive definite matrix, the trace $\operatorname{tr}(\mathbf{X}^{-1}\mathbf{Y})$ approaches ∞ as $\mathbf{Y}$ approaches the Satake boundary of H. Together with the lower bound $\operatorname{tr}(\mathbf{X}^{-1}\mathbf{Y}) \geq n$, this shows that the function attains a minimum.

Now, suppose that two distinct points $\mathbf{Y}_1, \mathbf{Y}_2 \in H$ attain the minimum of $\operatorname{tr}(\mathbf{X}^{-1}\mathbf{Y})$. Consider the matrix

$$\overline{\mathbf{Y}} = \frac{1}{2}(\mathbf{Y}_1 + \mathbf{Y}_2).$$

By the strict concavity of $\log\det(\mathbf{Y})$ on the positive cone of $\mathrm{Sym}_n(\mathbb{R})$,

$$\log\det(\overline{\mathbf{Y}}) > \frac{1}{2}(\log\det(\mathbf{Y}_1) + \log\det(\mathbf{Y}_2)) = 0,$$

so $\det(\overline{\mathbf{Y}}) > 1$. Let $\mathbf{Y}_0 = \overline{\mathbf{Y}}\det(\overline{\mathbf{Y}})^{-1/n}$. Then $\mathbf{Y}_0 \in H$, and

$$\mathrm{tr}(\mathbf{X}^{-1}\mathbf{Y}_0) = \mathrm{tr}(\mathbf{X}^{-1}\mathbf{Y}_1)\det(\overline{\mathbf{Y}})^{-1/n} < \mathrm{tr}(\mathbf{X}^{-1}\mathbf{Y}_1),$$

a contradiction. Therefore, there is a unique point $\mathbf{Y} \in H$ minimizing the Selberg invariant $\mathfrak{s}(\mathbf{X}, \mathbf{Y})$. □

This is an analogue of the nearest-point projection, with the usual distance function replaced by the Selberg invariant. The lemma implies that Selberg nearest-point projections to planes in $\mathcal{X}_n$ exist uniquely.

Author Contributions

Yukun Du carried out all research and wrote the manuscript.

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